CHONGQING FENGDU LFG POWER GENERATION PROJECT

Project title	Chongqing Fengdu LFG Power Generation Project
Project ID	5352
Crediting period	22-Aug-2023 to 21-Aug-2033
Original date of issue	31-Oct-2024
Most recent date of issue	31-Oct-2024
Version	1.0
VCS Standard Version	VCS Standard, v4.7
Prepared by	Kang Yi, Lin Keming, Li Shuhui, Cao Qin, Wang peiyao Climate Bridge (Shanghai)Ltd. Block B, Level 24, Jiangong Mansion, 33 Fushan Road, Pudong New Area, Shanghai, P.R. of China Telephone: +86 21 6246 2036 Email: kang.yi@climatebridge.com; projects@climatebridge.com http://www.climatebridge.com

CONTENTS

1 PROJECT DETAILS

1.1 Summary Description of the Project

Chongqing Fengdu LFG Power Generation Project (hereafter referred to as “the project”) is a landfill gas (LFG) recovery and utilization project developed by Fengdu Zhaoqian New Energy Technology Co., Ltd (hereinafter referred to as “the project proponent”) The project is located in the Fengdu County Municipal Solid Waste Landfill, Dali Village, Mingshan Town, Fengdu County, Chongqing City, People's Republic of China.

Fengdu County Municipal Solid Waste Landfill is managed by the Fengdu County Urban Administration Bureau. The landfill was put into operation in 2003, with a total storage capacity of 149 million cubic meters. The planned service life of the landfill area is 17.2 years, and the municipal solid waste (MSW) deposition rate is estimated to be 204 tons/day.

The purpose of the project is to capture and utilize the landfill gas for power generation. The electricity generated by the project is supplied to Southwest Power Grid (SWPG). The new gas extraction and utilization systems have been installed by the project. Before the implementation of the project activity, the landfill gas that captured and utilized by the project was emitted into the atmosphere directly. Equivalent electricity exported to the grid by the project was provided by Southwest Power Grid (SWPG) which is dominated by coal-fired power stations. This is the scenario that existed prior to the implementation of the project activity and is also the baseline scenario.

The project will reduce Greenhouse gas (GHG) emissions by avoiding CO_2 emissions from the fossil fuel-based power plants connected to the grid and by avoiding GHG emissions from releasing LFG into atmosphere at the landfill site.

The total installed capacity of the project is 2.134MW. The project is expected to supply a total of 47,746 MWh electricity to SWPG during the fixed 10-year crediting period (from 22-Aug-2023 to 21-Aug-2033), with the annual average amount of 4,775 MWh. The total amount of emission reductions is expected to be 189,104 tCO_2e during the fixed 10-year crediting period, and the average annual emission reduction is 18,910 tCO_2e.

1.2 Audit History

Audit type	Period	Program	Validation/verification body name	Number of years
Validation	NA	VCS	Carbon Check (India) Private Ltd.	NA

1.3 Sectoral Scope and Project Type

Sectoral scope	Sectoral Scope 1: Energy industries (renewable - /non-renewable sources) Sectoral Scope 13: Waste handling and disposal.
Project activity type	LFG Power Generation Project

1.4 Project Eligibility

1.4.1 General eligibility

The scope of the VCS Program includes:

1) The seven Kyoto Protocol greenhouse gases: The emissions reduction of the project comes from two sources: 1) Methane (CH_4) emissions as a result of the previously vented landfill gas that is captured and destroyed in the project scenario; 2) CO_2 emissions from the production of the equivalent amount of electricity replaced by the Project that would otherwise have been purchased from the fossil fuel fired power plants of Southwest Power Grid (SWPG). Thus, the project applicable to this scope.

2) Ozone-depleting substances: Not Applicable

3) Project activities supported by a methodology approved under the VCS Program through the methodology approval process: Not Applicable

4) Project activities supported by a methodology approved under a VCS approved GHG program, unless explicitly excluded under the terms of Verra approval: The methodology AMS-III.G (version 10.0) and AMS-I.D (version 18.0) of the project utilized are methodologies approved under CDM Program, which is a VCS approved GHG program. The methodologies have been proven not to be excluded by checking the Verra website[1]. Also, the Project is not a fragmented part of a larger project or activity, thus within the scale and/or capacity limits of the methodologies. No single cluster of project activity instances exceeds the capacity limit.

5) Jurisdictional REDD+ programs and nested REDD+ projects as set out in the VCS Program document Jurisdictional and Nested REDD+ (JNR) Requirements: Not Applicable

The project does not belong to projects that can reasonably be assumed to have generated GHG emissions primarily for the purpose of their subsequent reduction, removal, or destruction.

Furthermore, the project does not belong to the project activities excluded in Table 1 of VCS Standard 4.7.

According to requirements of VCS Standard 4.7, Non-AFOLU projects applying a standardized method shall complete validation within two years of the project start date. The validation

[1] https://verra.org/methodologies-main/

deadline of the Project is 21-Aug-2025 since it started operation on 22-Aug-2023. The Project meets the timeline criteria. In addition, The Project follows the requirements that the project is listed on the project pipeline with a status of under validation before the opening meeting. Furthermore, validation does not begin until the 30-day public comment period has begun.

Thus, the project is eligible under the scope of VCS program.

1.4.2 AFOLU project eligibility

Not applicable as the project is a non-AFOLU project.

1.4.3 Transfer project eligibility

Not applicable as the project is neither a transfer project nor CPAs seeking registration.

1.5 Project Design

☒ Single location or installation

☐ Multiple locations or project activity instances (but not a grouped project)

☐ Grouped project

1.5.1 Grouped project design

Not applicable as the project is not a grouped project.

1.6 Project Proponent

Organization name	Fengdu Zhaoqian New Energy Technology Co., Ltd
Contact person	Xie Handian
Title	Manager
Address	No. 11, Group 7, Dali Village, Mingshan Street, Fengdu County, Chongqing, China
Telephone	+86 021-23019950
Email	3542346576@qq.com

1.7 Other Entities Involved in the Project

Organization name	Climate Bridge (Shanghai) Ltd.
Role in the project	Consultancy
Contact person	Gao Zhiwen
Title	General Manager
Address	Block B, Level 24, Jiangong Mansion, 33 Fushan Road, Pudong New Area, Shanghai, China
Telephone	+86 021-62462036
Email	gao.zhiwen@climatebridge.com

1.8 Ownership

The project owner is Fengdu Zhaoqian New Energy Technology Co., Ltd. Who has the legal right to control and operate the project activity. The business license, Project approval issued by Chongqing Development and Reform Commission, the approval of Environmental Impact Assessment (EIA) issued by Fengdu Bureau of Ecology and Environment, Power Purchase Agreement (PPA) signed with the purchaser National Grid Chongqing Electric Power Co., Ltd. Fengdu County Power Supply Branch, the equipment purchasing contract, and the construction contract are evidence of the ownership of the project and carbon credits generated.

1.9 Project Start Date

Project start date	22-Aug-2023
Justification	According to VCS standard v4.7, the project start date is the date on which the project began generating GHG emission reductions or removals. The power generator was put into operation on 22-Aug-2023, which is the earliest date that the project started generating emission reductions.

1.10 Project Crediting Period

Crediting period	☐ Seven years, twice renewable ☒ Ten years, fixed ☐ Other (state the selected crediting period and justify how it conforms with the VCS Program requirements)

Start and end date of first or fixed crediting period	22-Aug-2023 to 21-Aug-2033

1.11 Project Scale and Estimated GHG Emission Reductions or Removals

☒ < 300,000 tCO_2e/year (project)

☐ ≥ 300,000 tCO_2e/year (large project)

Calendar year of crediting period	Estimated GHG emission reductions or removals (tCO_2e)
22-Aug-2023 to 31-Dec-2023	9,591
01-Jan-2024 to 31-Dec-2024	27,844
01-Jan-2025 to 31-Dec-2025	29,255
01-Jan-2026 to 31-Dec-2026	31,084
01-Jan-2027 to 31-Dec-2027	25,431
01-Jan-2028 to 31-Dec-2028	20,482
01-Jan-2029 to 31-Dec-2029	16,280
01-Jan-2030 to 31-Dec-2030	12,281
01-Jan-2031 to 31-Dec-2031	8,864
01-Jan-2032 to 31-Dec-2032	5,929
01-Jan-2033 to 21-Aug-2033	2,063
Total estimated ERRs during the first or fixed crediting period	189,104
Total number of years	10
Average annual ERRs	18,910

1.12 Description of the Project Activity

The Project activity has installed a comprehensive system for LFG recovery and utilization on the existing landfill site. The system consists of LFG collection, LFG pre-treatment and electricity generation. The key systems involved in the project are as follows:

- Gas collection system

The landfill gas collecting system is mainly composed of a gas gathering well, a gas gathering main pipe and gas collecting trunk pipes, and the basic process is as follows: the gas in the landfill flows to a specific gas collection well due to the pressure difference. The gas collecting well branch pipe leads the gas from the gas collecting well to the gas collecting trunk pipe, and the gas collecting trunk pipe then transports the gas to the gas collecting main pipe, and the gas is sent to the gas pre-treatment device from the gas collecting main pipe.

- **Gas pre-treatment system**

Before electricity generation, the LFG is pre-treated to remove impurities and moisture, etc., to prevent the corrosion of the electricity generation system. In addition, landfill gas should be in a continuously stable condition before it flows into the gas engine. The pre-treatment consists of pre-filtration, dehumidification, dewatering, cooling, and fine filtration.

- **Electricity generation system**

The electricity generated using LFG, except a small portion for on-site usage, is be sold to the SWPG through the nearby distribution network. Two gas generators with a rated capacity of 1.067 MW each have been installed by the project, and the total capacity is 2.134 MW. This is a well-known and high reliable technology for biogas utilization. High performance and reliability are guaranteed for the equipment.

The electricity generated by the project is transmitted to the 110kV Mingshan substation of SWPG through one 10kV transmission line. Electricity meter has been installed at the outlet of the onsite 10kV switchgear box 39 Mingzheng Line (connection point of the project and the power grid) to monitor the quantity of electricity supplied to SWPG and electricity imported from SWPG.

The detailed information of the Gas Generator of the project is shown in the table below.

Table 1.1 Main Equipment Parameter[2]

System	Parameter	Value	Unit
Gas collection system	Type	Integrated system with impermeable cover and vertical wells	-
	Capture efficiency	50%	-
Gas pre-treatment system	Type	The pretreatment system is composed of roots air unit, air-cooled radiator, filter, pneumatic shut-off valve, pipeline flame	--

[2] According to Equipment Purchase Contract

		arrester, measuring instrument, control cabinet and air compressor	
	Number	1	-
	Capacity	1500	Nm^3/h
Power generation system	Type	JGS320 GS-L.L	-
	Quantity	2	-
	Rated capacity	1.067	MW
	Rated power factor	1	-
	Rated frequency	50	Hz
	Rated Voltage	400	V
	Manufacturer	GE Jenbacher GmbH	-

1.13 Project Location

The Project Located in the Fengdu County Municipal Solid Waste Landfill, Dali Village, Mingshan Town, Fengdu County, Chongqing City, People's Republic of China. The coordinates of the project site are 29°53'33.184″N, 107°39′57.755″E. The project location is shown in Figure 1.1.

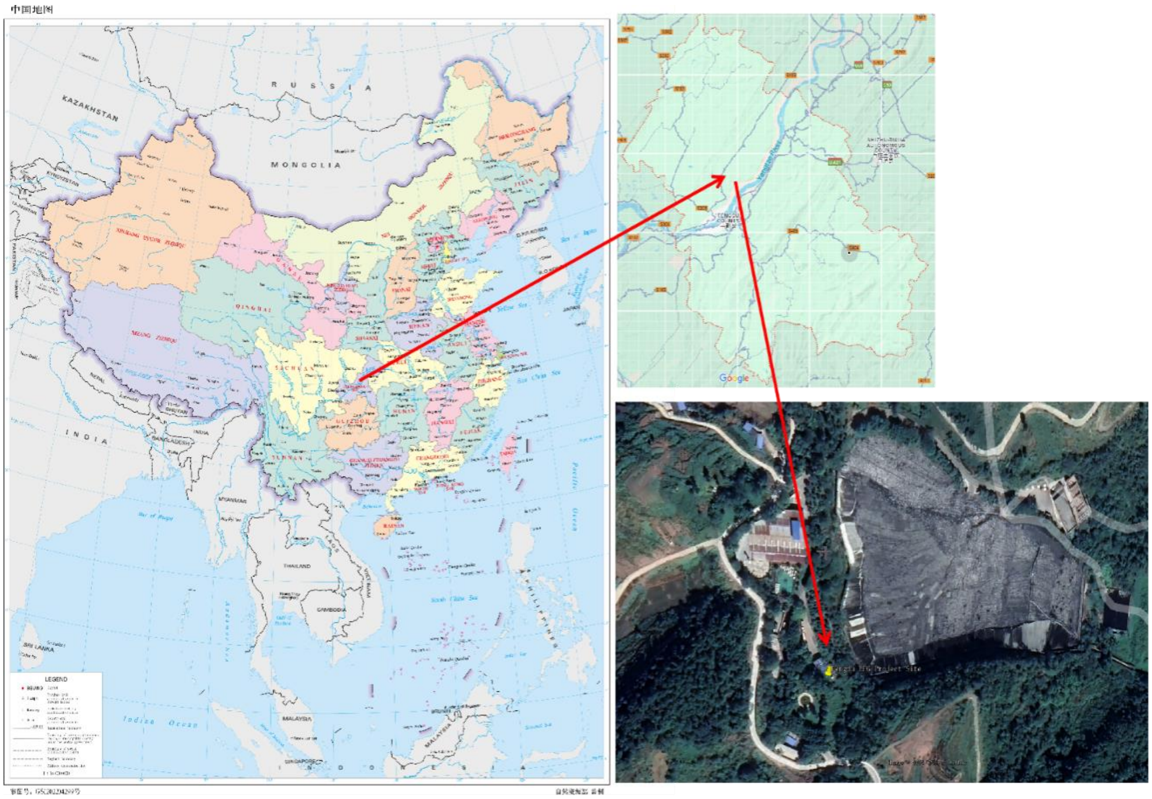

Figure 1.1 Location of the Project

1.14 Conditions Prior to Project Initiation

Prior to the implementation of the project activity, the LFG generated from Chongqing Fengdu Landfill was directly released into the atmosphere without capture or utilization; the equivalent amount of electricity supplied by the project activity was supplied by the fossil-fuel-dominated Southwest Power Grid (SWPG).

The baseline scenario is the same as the conditions existing prior to the project initiation. Please refer to Section 3.4 (Baseline Scenario).

1.15 Compliance with Laws, Statutes and Other Regulatory Frameworks

The Project complies with all Chinese relevant laws and regulations, as shown in Table 1.2.

Table 1.2 Compliance with relevant laws and regulations

Laws and regulations	The Project
Regulations on the Approval and Recordation of Enterprise Investment Projects[3], Administrative	The Project has obtained the project approval from Chongqing Municipal

3 http://www.gov.cn/zhengce/content/2016-12/14/content_5147959.htm

Measures on the Approval and Recordation of Enterprise Investment Projects[4], which sets out the procedures of project approval and recordation.	Development and Reform Commission, in compliance with the regulation and the administrative measure.
Law of People's Republic of China on Environmental Impact Assessment[5], which sets out the requirements on the completion and approval of the environmental impact assessment (EIA) report/form of construction projects.	The (EIA) form of the Project has been completed and then approved by Fengdu County Ecological Environment Bureau, which is following the provisions in the law.
Construction Law of the People's Republic of China[6], which sets out the requirements on application and approval of the construction permit prior to the project construction; Construction projects that have been approved for commencement reports in accordance with the authority and procedures stipulated by the State Council are no longer required to obtain construction permits.	The Project obtained the commencement report prior to the construction in compliance with the provisions in the law.
Catalogue for the Guidance of Industrial Structure Adjustment[7], which lists projects in three categories: encouragement category, restriction category and elimination category.	The Project is a landfill gas project and belongs to the encouragement category of the Catalogue.

1.16 Double Counting and Participation under Other GHG Programs

1.16.1 No Double Issuance

Is the project receiving or seeking credit for reductions and removals from a project activity under another GHG program?

☐ Yes ☒ No

1.16.2 Registration in Other GHG Programs

4 https://www.ndrc.gov.cn/xxgk/zcfb/fzggwl/201703/W020190905495074985482.pdf

5 https://www.mee.gov.cn/ywgz/fgbz/fl/201901/t20190111_689247.shtml

6http://www.npc.gov.cn/zgrdw/npc/xinwen/2019/05/07/content_2086833.htm?eqid=f69aa65f000cfe8500000004643a0ff9

7 http://www.gov.cn/xinwen/2019-11/06/5449193/files/26c9d25f713f4ed5b8dc51ae40ef37af.pdf

Has the project registered under any other GHG programs?

☐ Yes ☒ No

Is the project active under the other program?

☐ Yes ☒ No

1.16.3 Projects Rejected by Other GHG Programs

Has the project been rejected by any other GHG programs?

☐ Yes ☒ No

1.17 Double Claiming, Other Forms of Credit, and Scope 3 Emissions

1.17.1 No Double Claiming with Emissions Trading Programs or Binding Emission Limits

Are project reductions and removals or project activities also included in an emissions trading program or binding emission limit?

☐ Yes ☒ No

The project owner is not part of any emission trading program. The net GHG emission reductions from the project is not used for compliance with emission trading programs or to meet binding limits on GHG emissions. The project activity has not participated under any other GHG programs.

China has a national emissions trading scheme only cover the high-emission industries, such as thermal power generation, petrochemical, chemical, building materials, iron and steel, non-ferrous, paper, aviation and other key emission industries that emitted at least 26,000 tons of CO_2e/year[8]. And the project activity is not included the mandatory emission control scheme and there is no emission cap enforced for the project owner according to the enforced company list[9] in public information. Hence, it is confirmed that the emission reductions is not double counted.

1.17.2 No Double Claiming with Other Forms of Environmental Credit

Has the project activity sought, received, or is planning to receive credit from another GHG-related environmental credit system?

☐ Yes ☒ No

[8] https://www.mee.gov.cn/xxgk2018/xxgk/xxgk06/202302/t20230207_1015569.html

[9] https://sthjj.cq.gov.cn/zwgk_249/zfxxgkml/zcwj/qtwj/202405/t20240514_13206151.html

1.17.3 Supply Chain (Scope 3) Emissions

Do the project activities specified in Section 1.12 affect the emissions footprint of any product(s) (goods or services) that are part of a supply chain?

☐ Yes ☒ No

1.18 Sustainable Development Contributions

The Project contributes to sustainable development in the following ways:

- Increase electricity production from renewable sources, which certainly reduces the consumption of fossil fuel in Power Grid and further help to achieve the national action to promote renewable energy development. Thus, the project achieved SDG 7 Affordable and Clean Energy[10].
- The project provides job opportunities and increase tax revenue, which has a positive effect on the local economy. Thus, the project achieved SDG 8 Decent Work and Economic Growth[11].
- Reduce greenhouse gas emissions. The project recovers and destroys biogas that would otherwise be released directly into the atmosphere, effectively reducing greenhouse gas emissions, which contributes to China's commitment to peak carbon dioxide emissions before 2030. Thus, the project achieved SDG 13 Climate Action[12].

The project contributes to achieving Chinese sustainable development priorities is described as below:

SDG	Indicators	Chinese Sustainable Development Progress	Project activity contribution
	SDG 7 "Ensure access to affordable, reliable, sustainable and modern energy for all"	By the end of 2015, China had fully solved the problem of providing electricity to all people without electricity and reached the goal of the Sustainable Development Agenda 15 years ahead of schedule. From 2015 to 2020, the share of non-fossil	The project is to capture and utilize biogas and LFG for power generation and accordingly substitute equivalent electricity from thermal power. This contributes to

[10] https://sdgs.un.org/goals/goal7

[11] https://sdgs.un.org/goals/goal8

[12] https://sdgs.un.org/goals/goal13

		energy increased from 14.5 % to 19.6%, while the share of raw coal decreased from 72.2% to 67.6%.	achieve China's stated sustainable development priorities "By 2030, the proportion of non-fossil energy consumption will reach about 25%".
	SDG 8 "Promote sustained, inclusive and sustainable economic growth, full and productive employment and decent work for all".	China has deeply implemented the innovation-driven development strategy, and small, medium, and micro enterprises have developed rapidly. Adhering to the policy of giving priority to employment, the unemployment rate has remained at a low level. By coordinating epidemic prevention and control with economic and social development, China has become the only major economy to achieve positive growth in 2020 and has made positive contributions to global economic recovery.	The project activity can provide employment opportunities for local villagers. This contributes to one of China's actions for promoting sustainable developing: "by 2030, achieve full and productive employment and decent work for all women and men, including for young people and persons with disabilities, and equal pay for work of equal value".
	SDG 13 "Take urgent action to combat climate change and its impacts".	In 2020, China's energy consumption per unit of GDP was reduced by 24.4% compared with 2012; carbon dioxide emissions per unit of GDP was reduced by 18.8% compared with 2015 and 48.4% compared with 2005, all of which have already fulfilled China's commitment to the international community in 2020 ahead of schedule.	The project is to capture and utilize biogas and LFG for power generation to avoid landfill gas emissions and CO_2 emissions from the production of the equivalent amount of electricity replaced by the project that would otherwise have been purchased from the SWPG. This contributes to achieve China's stated sustainable development priorities "

			China's carbon dioxide emissions would strive to peak by 2030 and strive to achieve carbon neutrality by 2060 ".

1.19 Additional Information Relevant to the Project

1.19.1 Leakage Management

As per methodology AMS-I.D (Version 18.0), general guidance on leakage in biomass project activities shall be followed to quantify leakages pertaining to the use of biomass residues. The project activity does not involve utilizing biomass residue, no leakage emissions are produced.

As per methodology AMS-III.G (Version 10.0), if the methane recovery technology is equipment transferred from another activity, leakage effects are to be considered. The Project does not involve equipment transfer, so there are no leakage effects.

Therefore, leakage management does not apply to the Project.

1.19.2 Commercially Sensitive Information

There is no commercially sensitive information has been excluded from the public version of the project description.

1.19.3 Further Information

There is no additional relevant legislative, technical, economic, sectoral, social, environmental, geographic, site-specific and/or temporal information that may have a bearing on the eligibility of the project, the net GHG emission reductions or removals, or the quantification of the project's net GHG emission reductions or removals.

2 SAFEGUARDS AND STAKEHOLDER ENGAGEMENT

2.1 Stakeholder Engagement and Consultation

2.1.1 Stakeholder Identification

Stakeholder Identification	The stakeholder identification process for the project involved a step-by-step approach. Initially, the project owner performed a project scope analysis. This included evaluating the geographic boundary, potential environmental impacts, and the communities served by the project. Following this, the project owner engaged in a stakeholder mapping exercise, identifying individuals and organizations with a vested interest or potential influence over the project. Interviews and surveys were conducted with residents, businesses, and subject matter experts to gather various perspectives. In addition, the project owner reviewed regulatory standards and consulted with government entities to ensure compliance and to identify any legal stakeholders. The following people are considered as the stakeholders of the project during the implementation stage. ■ Residents of the nearby village. ■ Staff of the project. ■ Relevant administrative staff of the local government.
Legal or customary tenure/access rights	Applicable legal frameworks and existing LFG cooperative development agreements govern land use and access rights. The project activity does not involve alterations to the existing legal or customary tenure/access rights of stakeholders, indigenous people (IPs), local communities (LCs), or customary rights holders, as the land is already designated for industrial purposes.
Stakeholder diversity and changes over time	The social diversity mainly includes the differences of age, education, and occupations. The nearby residents are dominated by farmers, who also have a relatively low level of education. The worker of the plant is in the range of legal enforcement which is between 22 and 60/55 (male/female) years old. From the public recruitment

	information, workers of the project at least have accepted junior high school education. Experts and management personnel might be people who accept higher education or older. But it's not absolute. The economic sources of the stakeholders mainly come from farming, migrant work, individual enterprises and working in enterprises and governments. There are some disparities in income between different staff members. The project employs some local residents to solve some employment problems and increase their income. The stakeholders involved in the project are all Han, no minority ethnic groups are involved, and no obvious significate cultural diversity identified among Han people. The interactions between the stakeholder groups are mainly collaboratively, involves project information sharing and joint decision-making. Over time, the members of each group will adjust, but their groups do not change much in terms of society, economy, culture, etc.
Expected changes in well-being	The project is expected to improve the local environment, increase clean electricity supply. Besides, the project provides numerous employment options for residents, which will subsequently lead to a reduction in the number of impoverished households. By increasing their income, these job opportunities will enable individuals to connect with one another and engage in more social activities, ultimately fostering greater interaction within the community. As a result, the overall welfare of the local communities will witness a substantial improvement.
Location of stakeholders	Stakeholders in this project located at the project site and within a few kilometers of it.
Location of resources	The project does not involve the territories and resources, or customary access of the stakeholders own.

2.1.2 Stakeholder Consultation and Ongoing Communication

Date of stakeholder consultation	03-Apr-2023

Stakeholder engagement process	Prior to the Project construction, the project owner conducted local stakeholder consultation by distributing survey questionnaires in April 2023, to get informed of the stakeholders' opinion about the Project in various aspects. A total of 20 questionnaires were distributed and collected for the survey. The project proponent puts up public announcements on bulletin boards at the landfill and at neighbourhoods near the project site on various dates two weeks ahead of the upcoming consultation. The general information about the project, the date and the purpose of the questionnaire survey as well as the project proponent's phone number and email address are clearly conveyed via these public announcements and phone calls. During the consultation, the project proponent distributes questionnaires at a variety of places to collect the stakeholders' opinions about the project in various aspects: they go to the landfill to distribute questionnaires to employees; they go to the neighbourhoods nearby to distribute questionnaires to families living there; they also go to the office buildings of the local government and the Ecology and Environment Bureau. In addition, stakeholders are able to get the questionnaires at the locations communicated via public announcements and phone calls.
Consultation outcome	No negative feedback was received in public announcement and the local community expressed their approval for the project. The 20 questionnaires were distributed to local stakeholders, and all questionnaires have been recollected. All surveyed members of the public are aware of the construction of the project and believe that the construction of the project is beneficial or basically has no impact on their lives, and all in favor of the construction of the project under the premise of taking environmental protection measures to ensure that the emission standards are met.
Ongoing communication	The project owner has set up an on-going communication mechanism to regularly hold stakeholder meetings, distribute questionnaires and give the response to various stakeholders. Communications with Local stakeholders are being carried out at periodic intervals. Moreover, local stakeholders can also raise their concerns and opinions directly by making a phone call to the project owner.

Stakeholder input	During the project operation, feedback from stakeholders through surveys, phone calls, etc., regarding the project were collected and regularly reported to company management. Regarding environmental impact or safety issues, the relevant responsible personnel would be promptly notified for inspection and corrective action upon receiving the feedback, and an immediate report would be submitted to the company management. Since no complaints were collected during the past consultations, there were no necessary or appropriate updates to the project design during the reporting period.

2.1.3 Free Prior and Informed Consent

Obtaining consent	The project is supported and encouraged by the government. The project has obtained project approval and EIA approval from local government. Meanwhile, the project owner obtained the approval from the Landfill site through signing the cooperation Agreement on Landfill Gas Collecting and utilization. The project has maintained ongoing communication with the local residents and has secured permission and support for its construction and operation, Therefore, the project complies with the relevant VCS regulations for FPIC. There are no ongoing or unresolved conflicts for this project.
Outcome of FPIC	Local stakeholders are supportive of the construction of the project. The project was constructed with the consent of government departments at all levels. The project has not encroached on land, relocated people without consent, and forced physical or economic displacement. Most surveyed stakeholders think the project improves regional economic development without adverse environmental impact. The survey shows that the stakeholders are supportive of the project.

2.1.4 Grievance Redress Procedure

Development process	If any grievances are received during the project construction and operation, the project proponent will take on a series of reactions to resolve stakeholders' concerns. When stakeholders express their misgivings in the questionnaire survey, the project proponent will explain related issues including the corresponding measures on time. After collecting the questionnaires and analyzing the survey results, the project proponent will contact the stakeholders by

	phone calls to elaborate on the measures against the concerns raised within one week and ask for their feedback; the project proponent will also inform them how to contact the project and express opinions if they have any concerns in the future. Any grievances received by phone calls, emails, or direct communication will be explained within one week by phone calls or face-to-face conversations with the claimants. If updates of the Project are necessary, the project proponent will assess the problem seriously, formulate a scientific solution, timely update the progress with the complainants and ask for feedback until the grievance is resolved.
Grievance redress procedure	Firstly, once getting the complaints or grievances, project owner shall contact and discuss with relevant community or other stakeholders within 3 days. Then the specific staff of project owner should propose a solution and mediation which is performed by relevant government agency within a week based on all collected information. Thirdly, the project owner will address their complaints or grievances by relevant legal method such as arbitration and courts if necessary. Finally, the entire complaints or grievances redress process shall be dealt within 30 days.

2.1.5 Public Comments

This project will open for public comment during the public comment period.

Comments received	Actions taken
To be updated	To be updated

2.2 Risks to Stakeholders and the Environment

2.2.1 Management Experience

The project owner, Fengdu Zhaoqian New Energy Technology Co., Ltd has sufficient capacity and experience in project management, and the project has been running since Aug 2023 without any problem. The technicians of project owner have professional licenses and extensive hands-on experience, and regular training is conducted to further improve the skills of its staff. As the parent company, Shenzhen Zhonglan Huanneng Co. Ltd oversees a vast network of LFG power generation projects spread across multiple locations, such as Nanchang Zhonglan Huanneng Technical Service Co. Ltd. Maiyuan LFG Power Generation Project (VCS 2358), who can provide corresponding technical and management support and training to the project owner.

Thus, the management team has expertise and experience in implementing similar project activities and engaging communities.

2.2.2 Risk Assessment

	Risks identified	Mitigation or preventative measure(s) taken
Natural and human-induced risks to stakeholders' wellbeing	No risk identified	Stakeholders' well-being is fully considered during the project operation. The Project is expected to improve the local environmental conditions by contributing to GHG reduction, providing a clean electricity supply, offering both long-term and temporary work opportunities, and bringing economic interest to the owner.
Risks to stakeholder participation	No risk identified	There were no risks identified including project design and consultation. The project was designed and implemented to avoid trade-offs including negative impacts on livelihoods and climate change adaptation. The mechanism for ongoing communication with local stakeholders has been set up.
Working conditions	No risk identified	The project has obtained project approval and EIA approval, in this sense, the project meets all the requirement of regulation and laws set up by Chinese government in aspect of environment, health and safety. According to EIA, the project owner has established and improved the safety production responsibility system; formulated safety production rules and regulations and operating procedures; formulated and implemented the safety production education and training plan. These measures ensure the effective

		implementation of safety production investment, eliminate safety hazards in time, and carry out the standardization construction of safety production.
Safety of women and girls	No risk identified	There were no risks identified related to the safety of women and girls in the local community due to project activities. The project pays attention to the occupational health of female employees, providing regular health checkups and mental health support. The pay structure is reviewed regularly to ensure fair pay for both male and female employees.
Safety of minority and marginalized groups, including children	No risk identified	There were no risks identified related to safety of minority and marginalized groups, including children. At the project design and planning stage, the special needs of children, minorities and marginalized groups are fully considered to ensure the inclusiveness of project facilities and services. An effective feedback mechanism was established to ensure that community members can give feedback on their opinions and problems at any time.
Pollutants (air, noise, discharges to water, generation of waste, and release of hazardous materials and chemical pesticides and fertilizers)	No risk identified	Measures have been identified in the EIA of the project to mitigate environmental pollution caused by construction and operation, which were implemented per environment pollution-related laws and regulations and supervised by the local government.

2.3 Respect for Human Rights and Equity

2.3.1 Labor and Work

	Risks identified	Mitigation or preventative measure(s) taken
Discrimination	No risk identified	Article 12 of Labour Law of the People's Republic of China enforces: Labourers, regardless of their ethnic group, race, sex, or religious belief, shall not be discriminated against in employment. The project owner strictly obeys the anti-discrimination rules set out in the law. No discrimination has occurred or will occur.
Sexual harassment	No risk identified	Article 40 of Law of People's Republic of China for the Protection of Women's Rights and Interests enforces: Sexual harassment of women is prohibited. Women victims have the right to complain to their units and relevant authorities. The project owner strictly obeys the anti-discrimination rules set out in the law. No discrimination or sexual harassment has occurred or will occur.
Equal pay for equal work	No risk identified	Article 46 of the Labour Law of the People's Republic of China stipulates that the distribution of wages shall be based on the principle of distribution according to work, with equal pay for equal work. The project owner strictly obeys the law.
Gender equity in labor and work	No risk identified	The project-employed women are provided equal pay for equal work. No discrimination on gender is allowed due to the HR policy. Safe and effective reporting channels and procedures for handling gender discrimination are easily accessible to all workers inclusive of staff and contracted workers employed by third parties.

Forced labor	No risk identified	Article 244 of the Criminal Law of the People's Republic of China [Crime of Forced Labor] Anyone who forces another person to work by means of violence, threat or restriction of personal freedom shall be sentenced to fixed-term imprisonment of not more than three years or detention, and shall be fined; if the circumstances are serious, he shall be sentenced to fixed-term imprisonment of not less than three years but not more than 10 years, and shall be fined. The project owner strictly obeys the law.
Child labor	No risk identified	Article 15 of the Labor Law of the People's Republic of China states: "Employers are prohibited from employing minors under the age of sixteen. The project owner strictly obeys the law.
Human trafficking	No risk identified	All employees inclusive of staff and contracted workers employed by third parties are employed voluntarily. HR policy of the project owner strictly follows the laws and regulations in China that human trafficking is prohibited.

2.3.2 Human Rights

Risks identified	Mitigation or preventative measure(s) taken
No risk identified	The project is committed to upholding the rights of LPs, LCs, and customary rights holders by engaging in ongoing, informed consultations, obtaining FPIC for any activities, and ensuring that their cultural heritage and traditional knowledge are respected and preserved by international human rights law, the United Nations Declaration on the Rights of Indigenous Peoples, and ILO Convention 169.

	The project has obtained approval from the local government in accordance with laws and regulations before project construction. In addition, the project owner has conducted regular local stakeholder consultations and all stakeholders have agreed and supported the construction and operation of the project. The project proponent ensures that all employees' labor contracts are fair and equitable, specifying working conditions, wages and working hours. Any form of deposit, withholding of identity documents or restriction of employees' freedom is prohibited, ensuring all employees are preserved by international human rights law.

2.3.3 Indigenous Peoples and Cultural Heritage

Risks identified	Mitigation(s) or preventative measure taken
No risk identified	As the territories and resources occupied by the project is within the area of Chongqing Fengdu Landfill owned and managed by local government, where no indigenous peoples and cultural heritage involved.

2.3.4 Property Rights

Risks identified	Mitigation or preventative measure(s) taken
No risk identified	The project prioritizes protecting stakeholders' property rights by engaging in regular consultations, complying with legal standards, documenting rights, offering accessible conflict resolution, and ensuring fair benefit sharing and transparent monitoring. The Project proponent has obtained land use rights before construction. The Project was built in the reserved area of the Fengdu landfill, which is state owned and managed by the local government, and the Chongqing Development and Reform Commission has approved the project construction. Therefore, the project activities do not involve any risks related to providing and preserving the property rights of indigenous people (LPs), local communities (LCs), or customary rights holders.

2.3.5 Benefit Sharing

Process used to design the benefit sharing plan	There is no considerable negative effect of project activities to any stakeholder groups. The project was implemented inside the Chongqing Fengdu Landfill which was managed by local government. The project has been approved by the local government. And by implementing the project, the local environment has been improved, such as odour from land fill, fire and explosion risk from LFG, etc. The project does not result in any risks related to protecting and preserving the property rights of IPs, LCs, and customary rights holders, and to protecting legal or customary tenure/access rights to territories, property, and resources. Therefore, no benefit sharing plan is required.
Summary of the benefit sharing plan	As clarified above, no benefit sharing plan is required for the project.
Approval and dissemination of benefit sharing plan	As clarified above, no benefit sharing plan is required for the project.

2.4 Ecosystem Health

	Risks identified	Mitigation or preventative measure(s) taken
Impacts on biodiversity and ecosystems	No risk identified	According to EIA of the project, there are no nature reserves in the project site, so the project has no further impact on biodiversity and ecosystems.
Soil degradation and soil erosion	No risk identified	The EIA report states that a soil impact assessment is not required for this type of project.
Water consumption and stress	No risk identified	According to the EIA report, the total water consumption of the project is 2.7 m3/d. Wastewater is mainly domestic sewage and condensate. Condensate wastewater is piped to the landfill. Domestic sewage is discharged into the landfill's septic

		tank, which is then treated by the landfill's evaporation pretreatment system until it reaches the standard.

2.4.1 Rare, Threatened, and Endangered Species

Is the project located in or adjacent to habitats for rare, threatened, or endangered species?

☐ Yes ☒ No

	Risks identified	Mitigation or preventative measure(s) taken
Habitats for rare, threatened, and endangered species	No risk identified	As per the EIA report, the project's location does not involve any habitats for rare, threatened, or endangered species.
Areas for habitat connectivity	No risk identified	The project is located in the area of Chongqing Fengdu Landfill, there is no areas for habitat connectivity.

2.4.2 Introduction of Species

Not applicable as the project does not plant or introduce any species.

2.4.3 Ecosystem Conversion

	Risks identified	Mitigation or preventative measure(s) taken
Ecosystem conversion	No risk identified	Not applicable as the project is a non-AFOLU project.

3 APPLICATION OF METHODOLOGY

3.1 Title and Reference of Methodology

Type (methodology, tool or module).	Reference ID, if applicable	Title	Version
Methodology	AMS-III.G.	Landfill methane recovery[13]	10.0
Methodology	AMS-I.D.	Grid connected renewable electricity generation[14]	18.0
Tool	TOOL04	Emissions from solid waste disposal sites[15]	08.1
Tool	TOOL05	Baseline, Project and/or Leakage Emissions from Electricity Consumption and Monitoring of Electricity Generation[16]	03.0
Tool	TOOL07	Tool to calculate the emission factor for an electricity system[17]	07.0
Tool	TOOL32	Positive lists of technologies[18]	04.0

3.2 Applicability of Methodology

The project satisfies all the applicability criteria of the methodology AMS-III.G (Version 10.0) and AMS-I.D (Version 18.0), of which the detailed description is listed in Table 3.1 below:

Table 3.1 Applicability of the methodology and tools

Methodology/Tool ID	Applicability condition	Justification of compliance

[13] https://cdm.unfccc.int/UserManagement/FileStorage/HN2W3BMY6RKUQOZDCVXS08F9LT1AI7

[14] https://cdm.unfccc.int/UserManagement/FileStorage/2P7FS6ZQAR84LG3NMKYUH50WI9ODBC

[15] https://cdm.unfccc.int/methodologies/PAmethodologies/tools/am-tool-04-v6.0.1.pdf

[16] https://cdm.unfccc.int/methodologies/PAmethodologies/tools/am-tool-05-v3.0.pdf

[17] https://cdm.unfccc.int/methodologies/PAmethodologies/tools/am-tool-07-v7.0.pdf

[18] https://cdm.unfccc.int/methodologies/PAmethodologies/tools/am-tool-32-v4.0.pdf

AMS-III.G	This methodology comprises measures to capture and combust methane from landfills (i.e. solid waste disposal sites) used for the disposal of residues from human activities including municipal, industrial, and other solid wastes containing biodegradable organic matter.	Applicable.
	Different options to utilize the recovered landfill gas as detailed in paragraph 4 of “AMS-III.H.: Methane recovery in wastewater treatment” (version 19.0) are eligible for use under this methodology. The relevant procedures in AMS-III.H. shall be followed in this regard.	Applicable. The project is to capture and utilize the landfill gas for electrical energy generation directly, which falls under 4(a) of para.4 of AMS-III.H. And the LFG power generation component of the project shall apply methodology AMS-I.D.
	Measures are limited to those that result in aggregate emission reductions of less than or equal to 60 kt CO2 equivalent annually from all Type III components of the project activity.	Applicable. Expected aggregate emission reductions from all type III components under the project activity are less than 60 kt CO2 equivalent annually. The project’s maximal methane emission reduction within a year is 18,910 tCO2e, which is less than 60,000 tCO2e.
	The proposed project activity does not reduce the amount of organic waste that would have been recycled in the absence of the project activity.	Applicable. The project does not impact the management of the landfill, also does not reduce the amount of organic waste that would have been recycled in the absence of the project activity.
	This methodology is not applicable if the management of the solid waste disposal	N.A.

	site (SWDS) in the project activity is deliberately changed in order to increase methane generation compared to the situation prior to the implementation of the project activity (e.g. other than to meet a technical or regulatory requirement). Such changes may include, for example, the addition of liquids to an SWDS, pre-treating waste to seed it with bacteria for the purpose of increasing the rate of anaerobic degradation of the SWDS, or changing the shape of the SWDS to increase methane production.	The management of the SWDS in the project activity will not be deliberately changed in order to increase methane generation compared to the situation prior to the implementation of the project activity.
AMS-I.D	This methodology is applicable to project activities that: (a)Install a Greenfield plant; (b)Involve a capacity addition in (an) existing plant(s); (c)Involve a retrofit of (an) existing plant(s); (d)Involve a rehabilitation of (an) existing plant(s)/unit(s); or Involve a replacement of (an) existing plant(s).	Applicable. The project installs a Greenfield LFG power plant at the existing Fengdu Landfill where there was no renewable energy power plant operating prior to the implementation of the project activity.
	Hydropower plants with reservoirs that satisfy at least one of the following conditions are eligible to apply this methodology (a) The project activity is implemented in an existing reservoir with no change in the volume of reservoir; (b) The project activity is implemented in an existing reservoir, where the volume of reservoir is increased and the power density of the project activity, as per definitions given in the project	N.A. The project is not a hydropower plant.

	emissions section, is greater than 4 W/m^2; (c) The project activity results in new reservoirs and the power density of the power plant, as per definitions given in the project emissions section, is greater than 4 W/m^2.	
	If the new unit has both renewable and non-renewable components (e.g., a wind/diesel unit), the eligibility limit of 15 MW for a small-scale CDM project activity applies only to the renewable component. If the new unit co-fires fossil fuel 5, the capacity of the entire unit shall not exceed the limit of 15 MW.	N.A. The new unit has only renewable components, and the installed capacity of the gas engines is 2.134 MW, significantly less than 15 MW.
	Combined heat and power (co-generation) systems are not eligible under this category.	N.A. The project does not involve a co-generation system.
	In the case of project activities that involve the capacity addition of renewable energy generation units at an existing renewable power generation facility, the added capacity of the units added by the project should be lower than 15 MW and should be physically distinct from the existing units.	N.A. The project does not involve the capacity addition at an existing renewable power generation facility.
	In the case of retrofit, rehabilitation, or replacement, to qualify as a small-scale project, the total output of the retrofitted, rehabilitated or replacement power plant/unit shall not exceed the limit of 15 MW.	N.A. The project does not involve retrofit, rehabilitation, or replacement
	In the case of landfill gas, waste gas, wastewater treatment, and agro-industries projects recovered methane emissions are eligible under a relevant Type III category. If the recovered methane is used for electricity	Applicable. The project recovers LFG for electricity generation. Hence recovered methane emissions are eligible under Type III

	generation for supply to a grid then the baseline for the electricity component shall be in accordance with the procedure prescribed under this methodology. If the recovered methane is used for heat generation or cogeneration other applicable Type-I methodologies such as "AMS-I.C.: Thermal energy production with or without electricity" shall be explored.	methodology AMS-III.G. And the baseline for the electrical component is in accordance with the procedure prescribed in AMS-I.D.
	In case biomass is sourced from dedicated plantations, the applicability criteria in the tool "Project emissions from cultivation of biomass" shall apply	N.A. The project does not involve biomass sourced from dedicated plantations.
TOOL04 Emissions from solid waste disposal sites (version 08.1)	The tool can be used to determine emissions for the following types of applications: Application A: The CDM project activity mitigates methane emissions from a specific existing SWDS. Methane emissions are mitigated by capturing and flaring or combusting the methane (e.g. "ACM0001: Flaring or use of landfill gas"). The methane is generated from waste disposed of in the past, including prior to the start of the CDM project activity. In these cases, the tool is only applied for an ex-ante estimation of emissions in the project design document (CDM-PDD). The emissions will then be monitored during the crediting period using the applicable approaches in the relevant methodologies (e.g. measuring the amount of methane captured from the SWDS); Application B: The CDM project activity avoids or involves the disposal of waste at an SWDS. An example of this	Applicable. The project is the condition of application A, that mitigates methane emissions from a specific existing SWDS by capturing and combusting for power.

	application of the tool is ACM0022, in which municipal solid waste (MSW) is treated with an alternative option, such as composting or anaerobic digestion, and is then prevented from being disposed of in an SWDS. The methane is generated from waste disposed or avoided from disposal during the crediting period. In these cases, the tool can be applied for both ex-ante and ex-post estimation of emissions. These project activities may apply the simplified approach detailed in 0 when calculating baseline emissions.	
	These two types of applications are referred to in the tool for determining parameters.	Applicable.
	In the case that: (a) different types of residual waste are disposed or prevented from disposal; or that (b) both MSW and residual waste(s) are prevented from disposal, then the tool should be applied separately to each residual waste and to the MSW.	The project only involves MSW disposed in Fengdu landfill site, doesn't involve disposal or preventing from disposal of residual waste.
TOOL05 Baseline, Project and/or Leakage Emissions from Electricity Consumption and Monitoring of Electricity Generation	If emissions are calculated for electricity consumption, the tool is only applicable if one out of the following three scenarios applies to the sources of electricity consumption: (a) Scenario A: Electricity consumption from the grid. The electricity is purchased from the grid only, and either no captive power plant(s) is/are installed at the site of electricity consumption or, if any captive power plant exists on site, it is either not operating or it is not physically able to provide electricity to the electricity consumer; (b) Scenario B: Electricity	Applicable. Applicable. The Project meets the requirement of Scenario A that the electricity consumption is from the grid.

	consumption from (an) off-grid fossil fuel fired captive power plant(s). One or more fossil fuel fired captive power plants are installed at the site of the electricity consumer and supply the consumer with electricity. The captive power plant(s) is/are not connected to the electricity grid; or (c) Scenario C: Electricity consumption from the grid and (a) fossil fuel fired captive power plant(s). One or more fossil fuel fired captive power plants operate at the site of the electricity consumer. The captive power plant(s) can provide electricity to the electricity consumer. The captive power plant(s) is/are also connected to the electricity grid. Hence, the electricity consumer can be provided with electricity from the captive power plant(s) and the grid.	
	This tool can be referred to in methodologies to provide procedures to monitor amount of electricity generated in the project scenario, only if one out of the following three project scenarios applies to the recipient of the electricity generated: (a) Scenario I: Electricity is supplied to the grid; (b) Scenario II: Electricity is supplied to consumers/electricity consuming facilities; or (c) Scenario III: Electricity is supplied to the grid and consumers/electricity consuming facilities.	Applicable. The Project meets the requirement of Scenario A and supply electricity to the grid - SWPG.
	This tool is not applicable in cases where captive renewable power generation technologies are installed to provide electricity in the project activity, in the baseline scenario or to sources of	Applicable. There were no captive renewable power generation technologies installed to provide electricity in the project activity, in the baseline scenario or to sources

	leakage. The tool only accounts for CO2 emissions.	of leakage. Only CO2 emissions are included in the Project.
TOOL07 Tool to calculate the emission factor for an electricity system (version 07.0)	This tool may be applied to estimate the OM, BM and/or CM when calculating baseline emissions for a project activity that substitutes grid electricity that is where a project activity supplies electricity to a grid or a project activity that results in savings of electricity that would have been provided by the grid (e.g. demand-side energy efficiency projects).	Applicable. The electricity generated by the Project is supplied to the grid - SWPG.
	Under this tool, the emission factor for the project electricity system can be calculated either for grid power plants only or, as an option, can include off-grid power plants. In the latter case, two sub-options under step 2 of the tool are available to the project participants, i.e. option II a and option II b. If option II a is chosen, the conditions specified in “Appendix 1: Procedures related to off-grid power generation” should be met. Namely, the total capacity of off-grid power plants (in MW) should be at least 10 percent of the total capacity of grid power plants in the electricity system; or the total electricity generation by off-grid power plants (in MWh) should be at least 10 percent of the total electricity generation by grid power plants in the electricity system; and that factors which negatively affect the reliability and stability of the grid are primarily due to constraints in generation and not to other aspects such as transmission capacity.	In the host country as off-grid power generation is not significant. Therefore, the emission factor for the project electricity system is calculated only for the grid power plants. Thus, this applicability criterion is satisfied.
	In the case of CDM projects, the tool is not applicable if the project electricity	Applicable.

	system is located partially or totally in an Annex I country.	The project electricity system is located in a non-Annex I country.
	Under this tool, the value applied to the CO_2 emission factor of biofuels is zero.	Applicable. The value applied to the CO_2 emission factor of biofuels is zero when calculating baseline emission factor for SWPG
TOOL32 Positive lists of technologies (version 04.0)	The use of this methodological tool is not mandatory for the project participants of a CDM project activity or CDM PoA for demonstrating their additionality.	Applicable The project selects to use the tool to demonstrate the additionality.
	This methodological tool shall be applied in conjunction with a small-scale or large-scale methodology which refers to this tool.	Applicable Applicable. The methodology AMS-III.G. (version 10.0) refers to the methodological tool.
	The positive lists as contained in section 5 of this tool are valid up to 10 March 2025. Notwithstanding the provisions on the validity of new, revised and previous versions of methodologies and methodological tools in the “Procedure: Development, revision and clarification of baseline and monitoring methodologies and methodological tools”, there will be no grace period for the application of this tool and the validity of the positive list after this date, including in cases where further technologies are added to the positive list through revisions of this tool before this date.	Applicable The validation is expected to be finished before March 2025. Thus, the tool is valid and applicable for the Project.

3.3 Project Boundary

According to the methodology AMS-III.G, the project boundary is the physical, geographical site of the Landfill where the LFG is captured and destroyed/used, and as per AMS-I.D, the spatial extent of the project boundary includes the project power plant and all power plants connected physically to the electricity system that the project power plant is connected to.

The project boundary includes Chongqing Fengdu Landfill, the entire LFG related system (including the collection system, the pre-treatment system and the power generation system) and all power plants connected to SWPG

The diagram of the project boundary is shown in Figure 3.1.

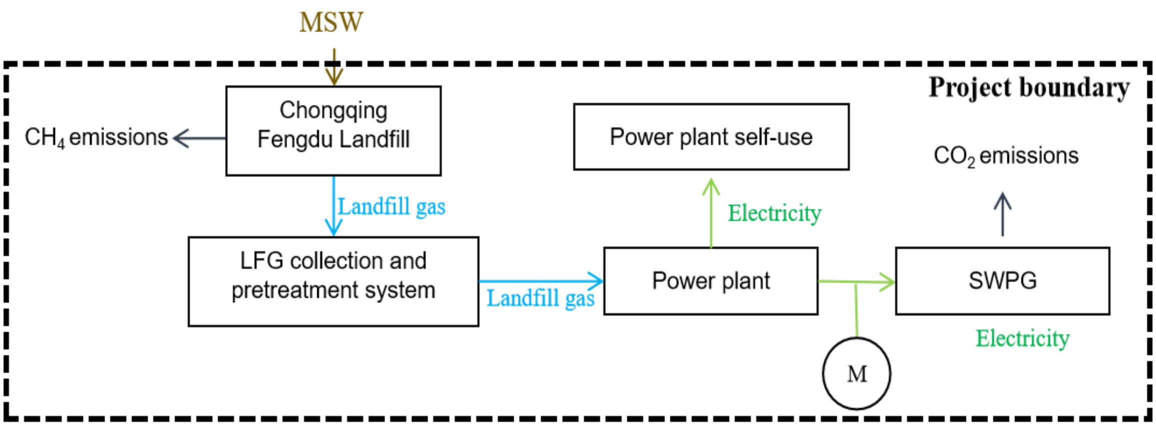

Figure 3.1 Diagram of the project boundary

The main emission sources and gases included in the project boundary are listed in table below:

Source		Gas	Included?	Justification/Explanation
Baseline	Emissions from decomposition of waste at the SWDS site	CO_2	No	CO_2 emissions from decomposition of organic waste are not accounted since the CO_2 is also released under the project activity.
		CH_4	Yes	The major source of emissions in the baseline.
		N_2O	No	N_2O emissions are small compared to CH_4 emissions from SWDS. This is conservative.
	Emissions from electricity generation	CO_2	Yes	Major emission source, given that power generation is included in the project activity.
		CH_4	No	Excluded for simplification. This is conservative.
		N_2O	No	Excluded for simplification. This is conservative.
	Emissions from heat generation	CO_2	No	The project activity does not involve heat generation.
		CH_4	No	

Source		Gas	Included?	Justification/Explanation
		N_2O	No	
	Emissions from the use of natural gas	CO_2	No	The project activity does not involve the use of natural gas.
		CH_4	No	
		N_2O	No	
Project	Emissions from fossil fuel consumption for purposes other than electricity generation or transportation due to the project activity	CO_2	No	The project activity does not involve fossil fuel consumption for purposes other than electricity generation or transportation due to the project activity.
		CH_4	No	Excluded for simplification. This emission source is assumed to be very small.
		N_2O	No	
	Emissions from electricity consumption due to the project activity	CO_2	Yes	The project activity may use electricity from the SWPG, so CO2 may be an important emission source.
		CH_4	No	This emission source is assumed to be very small.
		N_2O	No	This emission source is assumed to be very small.
	Emissions from flaring	CO_2	No	Excluded for simplification.
		CH_4	No	
		N_2O	No	
	Emissions from distribution of LFG using trucks and dedicated pipelines	CO_2	No	The project activity does not involve distribution of LFG using trucks or dedicated pipelines.
		CH_4	No	
		N_2O	No	

3.4 Baseline Scenario

As per AMS-III.G, the baseline scenario is the situation where, in the absence of the project activity, biomass and other organic matter are left to decay within the project boundary, and methane is emitted to the atmosphere, possibly with capture of LFG and destruction through flaring to comply with regulations or contractual requirements.

As per AMS-I.D, the baseline scenario is that the electricity delivered to the grid by the project activity would have otherwise been generated by the operation of grid-connected power plants and by the addition of new generation sources into the grid.

Therefore, the baseline scenario for the project activity is in the absence of the project activity, LFG from Fengdu Landfill site is emitted directly to the atmosphere which complies with all national and local regulations, and the equivalent amount of electricity supplied by the Project would have been supplied by SWPG.

3.5 Additionality

The project uses the following steps to demonstrate additionality. The project meets the additionality criteria as outlined in the VCS methodology as:

Step 1: Regulatory Surplus: the project is not mandated by any legal requirements. Please refer to section 3.5.1 below for details.

Step 2: Positive List: the project conforms to the positive list as stipulated in the applies the simplified procedures in Section 5.2 of the methodology AMS-III.G (version 10.0) and CDM "TOOL32: Positive lists of technologies" (version 04.0). Please refer to section 3.5.2 below for details.

3.5.1 Regulatory Surplus

Is the project located in an UNFCCC Annex 1 or Non-Annex 1 country?

☐ Annex 1 country ☒ Non-Annex 1 country

Are the project activities mandated by any law, statute, or other regulatory framework?

☐ Yes ☒ No

If the project is located inside a Non-Annex 1 country and the project activities are mandated by a law, statute, or other regulatory framework, are such laws, statutes, or regulatory frameworks systematically enforced?

☐ Yes ☒ No

According to the relevant applicable laws and regulations at the time of submission of registration, i.e. Renewable Energy Law of the People's Republic of China which came into effect on 01-Jan-2006, Agenda for China's 14th Five-Year Plan (2021-2025), there are no new national and/or sectoral policies that could affect the baseline scenario during the crediting period. Therefore, the project is surplus to regulatory requirements.

3.5.2 Additionality Methods

The project activity exclusively applies the technology listed in Section 5.1.1 (landfill gas recovery and its gainful use) of TOOL32 (version 04.0) and it fulfils the related conditions as follow. Therefore, the project activity is deemed automatically additional. details as follows:

Section 5.1.1 of TOOL32 (version 04.0)	The Project
Landfill gas recovery and its gainful use The project activities and PoAs at new or existing landfills (greenfield or brownfield) are deemed automatically additional if it is demonstrated that prior to the implementation of the project activities and PoAs the landfill gas (LFG) was only vented and or flared (in the case of brownfield projects)or would have been only vented and/or flared(in the case of greenfield projects) but not utilized for energy generation, and that under the project activities and PoAs any of the following conditions are met: (a) The LFG is used to generate electricity in one or several power plants with a total nameplate capacity that equals or is below 10 MW; (b) The LFG is used to generate heat for internal or external consumption; (c) The LFG is flared.	The Project involves landfill gas recovery and its gainful use. Prior to the implementation of the project activity, the LFG captured and utilized by the project from Chongqing Fengdu Landfill was vented to the atmosphere but not utilized for energy generation. The project activity utilizes LFG to generate electricity and the total nameplate capacity of all generators installed in the project is 2.134 MW, which is below 10 MW.

The additionality of the Project has thus been demonstrated.

3.6 Methodology Deviations

There is no methodology deviation for the project. Describe and justify any methodology

4 QUANTIFICATION OF ESTIMATED GHG EMISSION REDUCTIONS AND REMOVALS

4.1 Baseline Emissions

As per the methodology AMS-III.G (Version 10.0) and AMS-I.D (Version 18.0), the baseline emissions are determined according to Equation (1) and comprise the following sources:

Methane emissions from the SWDS in the absence of the project activity.

Electricity generation using fossil fuels or supplied by the grid in the absence of the project activity.

$$BE_y = BE_{CH4,y} + BE_{EC,y} \tag{1}$$

Where:

BE_y	=	Baseline emissions in year *y* (tCO_2e/yr)
$BE_{CH4,y}$	=	Baseline emissions of methane from the SWDS in year *y* (tCO_2e/yr)
$BE_{EC,y}$	=	Baseline emissions associated with electricity generation in year *y* (tCO_2/yr)

Based on AMS-I.D (version 18.0), the baseline emissions associated with electricity generation are to be calculated as below:

$$BE_{EC,y} = EG_{PJ,y} \times EF_{grid,y} \tag{2}$$

Where:

$BE_{EC,y}$	=	Baseline emissions associated with electricity generation in year *y* (t CO_2)
$EG_{PJ,y}$	=	Quantity of net electricity generation that is produced and fed into the grid as a result of the implementation of the CDM project activity in year *y* (MWh)
$EF_{grid,y}$	=	Combined margin CO_2 emission factor for grid connected power generation in year *y* calculated using the latest version of the "Tool to calculate the emission factor for an electricity system" (t CO_2/MWh)

For Greenfield power plants,

$$EG_{PJ,y} = EG_{PJ,facility,y} \tag{3}$$

Where:

$EG_{PJ,facility,y}$ = Quantity of net electricity generation supplied by the project plant/unit to the grid in year *y* (MWh)

The value of $EF_{grid,y}$ is determined as 0.32965 tCO_2/MWh on the basis of Tool7: "Tool to Calculate the Emission Factor for an Electricity System" (Version 07.0), referring to Appendix 2 for the procedure.

As per AMS-III.G (Version 10.0), Baseline emissions of methane from the SWDS ($BE_{CH4,y}$) is calculated ex-ante using equation (2) below:

$$BE_{CH4,y} = \eta_{PJ} \times BE_{CH4,SWDS,y} - (1 - OX) \times F_{CH4,BL,y} \times GWP_{CH4} \quad (4)$$

Where:

$BE_{CH4,y}$ = Baseline emissions of methane from the SWDS in year *y* (t CO_2e/yr)

$BE_{CH4,SWDS,y}$ = Methane emission potential of a solid waste disposal site (in tCO_2e) calculated using the methodological tool "Emissions from solid waste disposal sites". This tool may be used:

- With the factor "*f* =0.0" because the amount of LFG that would have been captured and destroyed is already accounted for in this equation;
- With the definition of year x as 'the year since the landfill started receiving wastes, x runs from the first year of landfill operation (x=1) to the year for which emissions are calculated (x=y)'.

The amount of waste type j deposited each year x($W_{j,x}$)shall be determined by sampling (as specified in the above-mentioned tool), in the case that waste is generated during the crediting period. Alternatively, for existing SWDS, if the pre-existing amount and composition of the wastes in the landfill are unknown, they can be estimated by using parameters related to the serviced population or industrial activity, or by comparison with other landfills with similar conditions at regional or national level

OX = Oxidation factor (reflecting the amount of methane from SWDS that is oxidised in the soil or other material covering the waste) (dimensionless). A default value of 0.1 may be used

η_{PJ} = Efficiency of the LFG capture system that will be installed in the project activity. It is used for ex-ante estimation only. A default value of 50 percent may be used

$F_{CH4,BL,y}$ = Amount of methane in the LFG that would be flared in the baseline in year y (t CH_4/yr)

GWP_{CH4} = Global warming potential of CH_4 (t CO_2e/t CH_4)

$BE_{CH4,SWDS,y}$ is determined using the methodological tool "Emissions from solid waste disposal sites" (version 08.1). The following guidance should be taken into account when applying the tool:

(a) f_y in the tool shall be assigned a value of 0 because there's no LFG recovery and utilization facility in the baseline.

(b) In the tool, x begins with the year that the SWDS started receiving wastes (e.g. the first year of SWDS operation); and

(c) Sampling to determine the fractions of different waste types is not necessary because the waste composition can be obtained from previous studies.

The $BE_{CH4,SWDS,y}$ is calculated as follows:

$$BE_{CH4,SWDS,y} = \varphi_y \times (1 - OX) \times \frac{16}{12} \times F \times DOC_{f,y} \times MCF_y \times \sum_{x=1}^{y} \sum_{j} \left(W_{j,x} \times DOC_j \times e^{-k_j \times (y-x)} \times \left(1 - e^{-k_j}\right) \right) \qquad (5)$$

Where:

$BE_{CH4,SWDS,y}$	=	Baseline methane emissions occurring in year y generated from waste disposal at an SWDS during a time period ending in year y (tCO_2e/yr)
x	=	Years in the time period in which waste is disposed at the SWDS, extending from the first year in the time period (x = 1) to year y (x = y)
y	=	Year of the crediting period for which methane emissions are calculated (y is a consecutive period of 12 months)
$DOC_{f,y}$	=	Fraction of degradable organic carbon (DOC) that decomposes under the specific conditions occurring in the SWDS for year y (weight fraction)
$W_{j,x}$	=	Amount of solid waste type j disposed or prevented from disposal in the SWDS in the year x (t)
φ_y	=	Model correction factor to account for model uncertainties for year y
f_y	=	Fraction of methane captured at the SWDS and flared, combusted or used in another manner that prevents the emissions of methane to the atmosphere in year y
GWP_{CH4}	=	Global Warming Potential of methane
OX	=	Oxidation factor (reflecting the amount of methane from SWDS that is oxidized in the soil or other material covering the waste)
F	=	Fraction of methane in the SWDS gas (volume fraction)
MCF_y	=	Methane correction factor for year y
DOC_j	=	Fraction of degradable organic carbon in the waste type j (weight fraction)
k	=	Decay rate for the waste type j (1/yr)

j = Type of residual waste or types of waste in the MSW

The model correction factor (φ_y) depends on the uncertainty of the parameters used in the FOD model. For baseline emissions, project participants may choose between the following two options to calculate φ_y.

- **Option 1:** Use a default value. Use a default value: φ_y= $\varphi_{default}$. Default values for different applications and climatic conditions are provided in the section "Data and parameters not monitored" below.
- **Option 2:** Determine φ_y based on specific situation of the project activity

In the Project, option 1 is chosen to determine φ_y as 0.75 for both humid/wet and dry conditions.

According to the previous study in the Feasibility Study Report (FSR) of the Project and Tool04: "Emissions from solid waste disposal sites" (Version 08.1), the data of $DOC_{f,y}$, $W_{j,x}$, and k_j are determined as follow:

Table 4.1 Parameters of different waste types

j	a	b	c	d	e
	Wood	Paper	Food	Textile	Other (rubber, plastic, glass, metal, etc.)
DOC_j	43%	40%	15%	24%	0
k_j (1/yr)	0.03	0.06	0.185	0.06	0

For the Project meeting the requirements of Application A, the MCF should be selected as a default value (MCF_y = $MCF_{default}$) provided in the section "Data and parameters not monitored", where $MCF_{default}$ = 1 for anaerobic managed solid waste disposal sites.

Before the implementation of the Project, there's no LFG recovery and utilization facility in the baseline, so the f_y in the tool shall be assigned a value of 0. And since the start year of the landfill accepting wastes is 2003, x equals to 2003. According to Tool04: "Emissions from solid waste disposal sites" (Version 08.1), default OX is equal to 0.1, default $DOC_{f,y}$ is determined as 0.5, and the value of fraction of methane in the SWDS gas (volume fraction) (F) is 0.5. Based on IPCC Fifth Assessment Report (AR5), the value of GWP_{CH4} is equal to 28[19].

The determination of $F_{CH4,BL,y}$ follows a procedure to determine the amount of methane that would have been captured and destroyed (by flaring) in the baseline due to regulatory or contractual requirements, to address safety and odour concerns, or for other reasons (collectively

[19] Microsoft Word - Global-Warming-Potential-Values.docx (ghgprotocol.org)

referred to as requirement in this section). The four cases in Table 4-2 are distinguished. The appropriate case should be identified, and the corresponding instructions followed:

Table 4.2 Case for determining methane captured and destroyed in the baseline

Situation at the start of the project activity	Requirement to destroy methane	Existing LFG capture and destruction system
Case 1	No	No
Case 2	Yes	No
Case 3	No	Yes
Case 4	Yes	Yes

According to "Technical code for municipal solid waste sanitary landfill" (GB 50869-2013) issued by Ministry of Housing and Urban-Rural Development of the People's Republic of China and General Administration of Quality Supervision, Inspection and Quarantine of the People's Republic of China on 08-Aug-2013, LFG recovery and destruction is recommended but not mandated in China, i.e., no mandatory requirement to destroy methane. However, to be conservative, the project chooses case 2.

In this situation:

$$F_{CH4,BL,y} = F_{CH4,BL,R,y} \quad (6)$$

Where:

$F_{CH4,BL,R,y}$ = Amount of methane in the LFG which is flared in the baseline due to a requirement in year y (t CH_4/yr)

$F_{CH4,BL,R,y}$ should be determined based on the information contained in the requirement to destroy methane, as follows:

- If the requirement specifies the amount of methane that must be flared then that amount is $F_{CH4,BL,R,y}$;
- If the requirement specifies a percentage of the captured LFG that is required to be flared, the amount shall be calculated as follows:

$$F_{CH4,BL,R,y} = \rho_{reg,y} \times F_{CH4,PJ,capt,y} \quad (7)$$

Where:

$\rho_{reg,y}$ = Fraction of LFG that is required to be flared due to a requirement in year *y*

$F_{CH4,PJ,capt,y}$ = Amount of methane in the LFG which is captured in the project activity in year y (t CH_4/yr)

Project participants may choose to calculate $F_{CH4,PJ,capt,y}$ by either of the two options:

- **Option 1:** Calculate using the "Tool to determine the mass flow of a greenhouse gas in a gaseous stream", applying the following requirements:

(i) The gaseous stream tool shall be applied to the LFG pipeline immediately downstream of the LFG capture system and before any split in the gaseous flow to different uses or flares;

(ii) CH_4 is the greenhouse gases for which the mass flow should be determined;

(iii) The simplification offered for calculating the molecular mass of the gaseous stream is valid; and

(iv) The mass flow should be calculated on an hourly basis for each hour *h* in year *y*

- Option 2: Calculate as the sum of the amount of methane that is sent to the flare, electricity generating or heat generating equipment in year *y* as measured in section 5.4.1.1 in Methodology ACM0001, however, not taking into account the working hours of the equipment;

(i) If the requirement does not specify the amount or percentage of LFG that should be destroyed but requires the installation of a capture system, without requiring the captured LFG to be flared then:

$$F_{CH4,BL,R,y} = 0 \tag{8}$$

(ii) If the requirement does not specify any amount or percentage of LFG that should be destroyed, but requires the installation of a system to capture and flare the LFG, then a typical destruction rate of 20 per cent is assumed[20]:

$$F_{CH4,BL,R,y} = 0.2 \times F_{CH4,PJ,capt,y} \tag{9}$$

There is no requirement specifying the amount or percentage of LFG that should be destroyed, and no requirement on the installation of a system to capture and flare the LFG. As a result, it is assumed a typical destruction rate of 20 per cent by Option 2 (ii) to calculate the required LFG destruction.

An ex ante estimate of $F_{CH4,PJ,capt,y}$ is equal to amount of methane in the LFG which is flared and/or used in the project activity, $F_{CH4,PJ,y}$. It is required to estimate baseline emission of methane from the SWDS in order to estimate the emission reductions of the proposed project activity. It is determined as follows:

$$F_{CH4,PJ,y} = \eta_{PJ} \times BE_{CH4,SWDS,y}/GWP_{CH4} \tag{10}$$

Where:

$F_{CH4,PJ,y}$	=	Amount of methane in the LFG which is flared and/or used in the project activity in year *y* (t CH_4/yr)

[20] This default value of 20 per cent is based on assuming a situation in which: the efficiency of the LFG capture system in the project is 50 per cent; the efficiency of the LFG capture system in the baseline is 20 per cent; and, the amount captured in the baseline is flared using an open flare with a destruction efficiency of 50 per cent (consistent with the default value provided in the tool “Project emissions from flaring”). Project participants may propose and justify an alternative default value as a request for revision to this methodology.

η_{PJ}	=	Methane emission potential of a solid waste disposal site (in t CO_2e), calculated using the methodological tool "Emissions from solid waste disposal sites". This tool may be used: • With the factor "f=0.0" because the amount of LFG that would have been captured and destroyed is already accounted for in this equation; • With the definition of year x as 'the year since the landfill started receiving wastes, x runs from the first year of landfill operation (x=1) to the year for which emissions are calculated (x=y)'. The amount of waste type *j* deposited each year *x* ($W_{j,x}$) shall be determined by sampling (as specified in the above-mentioned tool), in the case that waste is generated during the crediting period. Alternatively, for existing SWDS, if the pre-existing amount and composition of the wastes in the landfill are unknown, they can be estimated by using parameters related to the serviced population or industrial activity, or by comparison with other landfills with similar conditions at regional or national level
$BE_{CH4,SWDS,y}$	=	Oxidation factor (reflecting the amount of methane from SWDS that is oxidised in the soil or other material covering the waste) (dimensionless). A default value of 0.1 may be used

4.2 Project Emissions

As per AMS-I.D. (Version 18.0), project emission of the project is zero.

As per the methodology AMS-III.G. (Version 10.0), project emissions are calculated as follows:

$$PE_y = PE_{power,y} + PE_{flare,y} + PE_{process,y} \tag{11}$$

Where:

PE_y	=	Project emissions in year *y* (t CO_2/yr)
$PE_{power,y}$	=	Emissions from the use of fossil fuel or electricity for the operation of the installed facilities in the year y (t CO_2/yr)
$PE_{flare,y}$	=	Emissions from flaring or combustion of the landfill gas stream in the year y (t CO_2/yr)
$PE_{process,y}$	=	Emissions from the landfill gas upgrading process in the year y (t CO_2e), determined by following the relevant procedures described in annex 1 of AMS-III.H.

The project does not involve the emissions from the consumption of fossil fuels, only emissions from the use of electricity for the operation of the installed facilities are involved. The project will consume electricity delivered from the grid when the project is not in operation. And this part of electricity has already been deducted from the quantity of electricity supplied by the project plant/unit to the grid when calculating the Quantity of net electricity generation supplied by the project plant/unit to the grid, which is then used to calculate the Baseline emissions associated with electricity generation ($BE_{EC,y}$) hence emissions from the use of electricity for the operation of the installed facilities are excluded, which means Emissions from the use of fossil fuel or electricity for the operation of the installed facilities equal to 0.

The project does not claim emission reductions from flaring, hence project emissions from flaring or combustion of the landfill gas stream are excluded.

The project does not involve a landfill gas upgrading process, hence the Emissions from the landfill gas upgrading process are also excluded.

In conclusion, the project emission of the project activity is 0.

4.3 Leakage Emissions

No leakage effects are accounted for under AMS-III.G. (Version 10.0) and AMS-I.D. (Version 18.0).

4.4 Estimated GHG Emission Reductions and Carbon Dioxide Removals

The emission reduction achieved by the project activity can be estimated ex-ante as follows.

$$ER_{y,estimated} = BE_y - PE_y - LE_y \quad (12)$$

Where:

$ER_{y,estimated}$	=	Emission reductions in year *y* (t CO_2e)
BE_y	=	Baseline emissions in year *y* (t CO_2e)
PE_y	=	Project emissions in year *y* (t CO_2e)
LE_y	=	Leakage emissions in year *y* (t CO_2e)

As per AMS-III.G. (Version 10.0), the actual emission reduction achieved for Type III component of the project (which means not including emission reduction from Type I component, i.e. LFG power generation) during the crediting period will be calculated ex post using the amount of methane recovered and destroyed/gainfully used by the project activity, calculated as:

$$ER_{y,calculated} = (1 - OX) \times \left(F_{CH4,PJ,y} - F_{CH4,BL,y}\right) \times GWP_{CH4} - PE_y - LE_y \quad (13)$$

Where:

$F_{CH4,PJ,y}$	=	Methane captured and destroyed/gainfully used by the project activity in the year *y* (t CH_4)

As per section 5.2 and 5.3 above, project emission equals to zero, and no leakage is considered for the project. Therefore, the total emission reduction by the project (both Type I and Type III components) is calculated as follows:

$$ER_{y,calculated} = (1 - OX) \times \left(F_{CH4,PJ,y} - F_{CH4,BL,y}\right) \times GWP_{CH4} + BE_{EC,y} \tag{14}$$

As specified above, to be conservative, $F_{CH4,BL,y} = 0.2 \times F_{CH4,PJ,capt,y}$.

For project activities that utilize the recovered methane for power generation, $F_{CH4,PJ,y}$ may be calculated as follows, based on the amount of monitored electricity generation, without monitoring methane flow and concentration:

$$F_{CH4,PJ,capt,y} = \frac{EG_y \times 3600}{NCV_{CH4} \times EE_y} \times D_{CH4} \tag{15}$$

Where:

EG_y	=	Electricity generation in year y (MWh)
3600	=	Conversion factor (1 MWh = 3600 MJ)
NCV_{CH4}	=	NCV of methane (MJ/Nm^3), use default value: 35.9 MJ/Nm^3
D_{CH4}	=	Density of methane at the temperature and pressure of the landfill gas in the year y (t/ m^3).
EE_y	=	Energy Conversion Efficiency of the project equipment determined from one of the following options: • Specification provided by the equipment manufacturer specifically for biogas fuel only if the equipment is designed to utilize biogas as fuel. If the specification provides a range of efficiency values, the highest value of the range shall be used for the calculation. • Default efficiency of 40 percent

- **Baseline emissions resulted from Solid Waste Disposal Site (AMS-III.G).**
  In the PD, Equation 4-6 are applied to ex-ante estimate the baseline emissions resulted from methane recovery:

Table 4.3 The ex-ante estimation of baseline emissions resulted from AMS-III.G.

Year	η_{PJ}	$BE_{CH4,SWDS,y}$ (tCO_2e)	OX	$F_{CH4,BL,y}$ (tCH_4)	GWP_{CH4} (tCO_2e/tCH_4)	$BE_{CH4,y}$ (tCO_2e)
2023	0.5	60,124.32	0.1	215	28	24,651

2024	0.5	63,904.41	0.1	242	28	25,854
2025	0.5	67,349.61	0.1	259	28	27,157
2026	0.5	71,475.93	0.1	273	28	28,859
2027	0.5	61,319.81	0.1	283	28	23,522
2028	0.5	52,785.90	0.1	300	28	18,839
2029	0.5	45,605.11	0.1	315	28	14,860
2030	0.5	39,553.49	0.1	346	28	11,051
2031	0.5	34,444.70	0.1	374	28	7,791
2032	0.5	30,123.61	0.1	400	28	4,992
2033	0.5	26,461.07	0.1	429	28	2,408
Total						**189983**

- **Calculation of estimated baseline emissions resulted from electricity generation:**

Baseline emissions resulting from AMS-I.D. are ex-ante estimated as follows:

Table 4.4 The ex-ante estimation of baseline emissions resulted from AMS-I.D

Year	$EG_{PJ,y}$ (MWh/y)	$EF_{grid,CM,y}$ (tCO_2/MWh)	$BE_{EC,y}$ (tCO_2)
2023	2,054.97	0.32965	677
2024	6,039.56	0.32965	1,990
2025	6,365.16	0.32965	2,098
2026	6,755.14	0.32965	2,226
2027	5,795.29	0.32965	1,910
2028	4,988.76	0.32965	1,644
2029	4,310.11	0.32965	1,420
2030	3,738.17	0.32965	1,232
2031	3,255.34	0.32965	1,073
2032	2,846.96	0.32965	938
2033	1,596.41	0.32965	526

Total	15734

- **Calculation of estimated emissions reductions:**

Considering the project emissions and leakage emissions are 0 as described above, the estimated emission reductions are calculated as follows:

Table 4.5 The ex-ante estimation of emissions reductions

Year	$BE_{CH4,y}$ (tCO_2e)	$BE_{EC,y}$ (tCO_2)	BE_y (tCO_2)	PE_y (tCO_2)	LE_y (tCO_2)	ER_y (tCO_2)
2023	8,914	677	9,591	0	0	9,591
2024	25,854	1,990	27,844	0	0	27,844
2025	27,157	2,098	29,255	0	0	29,255
2026	28,858	2,226	31,084	0	0	31,084
2027	23,521	1,910	25,431	0	0	25,431
2028	18,838	1,644	20,482	0	0	20,482
2029	14,860	1,420	16,280	0	0	16,280
2030	11,050	1,232	12,282	0	0	12,282
2031	7,790	1,073	8,863	0	0	8,863
2032	4,991	938	5,929	0	0	5,929
2033	1,537	526	2,063	0	0	2,063
Total	**173,370**	**15,734**	**189,104**	**0**	**0**	**189,104**

During the fixing 10-year crediting period from 22-Aug-2023 to 21-Aug-2033, the estimated emission reductions are calculated as follows:

Vintage period	Estimated baseline emissions (tCO_2e)	Estimated project emissions (tCO_2e)	Estimated leakage emissions (tCO_2e)	Estimated reduction VCUs (tCO_2e)	Estimated removal VCUs (tCO_2e)	Estimated total VCUs (tCO_2e)
22-Aug-2023 to 31-Dec-2023	9,591	0	0	9,591	0	9,591

01-May-2024 to 31-Dec-2024	27,844	0	0	27,844	0	27,844
01-Jan-2025 to 31-Dec-2025	29,255	0	0	29,255	0	29,255
01-Jan-2026 to 31-Dec-2026	31,084	0	0	31,084	0	31,084
01-Jan-2027 to 31-Dec-2027	25,431	0	0	25,431	0	25,431
01-Jan-2028 to 31-Dec-2028	20,482	0	0	20,482	0	20,482
01-Jan-2029 to 31-Dec-2029	16,280	0	0	16,280	0	16,280
01-Jan-2030 to 31-Dec-2030	12,282	0	0	12,281	0	12,282
01-Jan-2031 to 31-Dec-2031	8,863	0	0	8,864	0	8,863
01-Jan-2032 to 31-Dec-2032	5,929	0	0	5,929	0	5,929
01-Jan-2033 to 21-Aug-2033	2,063	0	0	2,063	0	2,063
Total	189,104	0	0	189,104	0	189,104

5 MONITORING

5.1 Data and Parameters Available at Validation

Data / Parameter	NCV_{CH4}
Data unit	MJ/Nm^3
Description	NCV of methane (MJ/Nm^3) use default value: 35.9 MJ/Nm^3
Source of data	AMS-III.G: “Landfill methane recovery” (Version 10.0)
Value applied	35.9 MJ/Nm^3
Justification of choice of data or description of measurement methods and procedures applied	-
Purpose of data	Calculation of baseline emissions
Comments	-

Data / Parameter	EE_y
Data unit	%
Description	Energy Conversion Efficiency of the project equipment
Source of data	AMS-III.G Landfill methane recovery (Version 10.0)
Value applied	40.0%
Justification of choice of data or description of measurement methods and procedures applied	EE_y shall be determined from one of the following options: • Specification provided by the equipment manufacturer specifically for biogas fuel only if the equipment is designed to utilize biogas as fuel. If the specification provides a range of efficiency values, the highest value of the range shall be used for the calculation • Default efficiency of 40 per cent The default value of 40 per cent is used for the Project, and the latter option is applicable.
Purpose of data	Calculation of baseline emissions
Comments	-

Data / Parameter	D_{CH4}
Data unit	tones/Nm^3
Description	Density of methane at the temperature and pressure of the landfill gas
Source of data	TOOL06: "Project emissions from Flaring" (Version 04.0)[21]
Value applied	0.000716 tones/Nm^3
Justification of choice of data or description of measurement methods and procedures applied	$LFG_{i,y}$ is reported at normal conditions of temperature and pressure, so the density of methane is also determined as 0.000716 tones/Nm^3 at normal conditions.
Purpose of data	Calculation of baseline emissions
Comments	-

Data / Parameter	GWP_{CH4}
Data unit	tCO_2e/tCH_4
Description	Global Warming Potential for methane
Source of data	IPCC AR5[22]
Value applied	28 tCO_2e/tCH_4
Justification of choice of data or description of measurement methods and procedures applied	-
Purpose of data	Calculation of baseline emissions
Comments	-

Data / Parameter	$EF_{grid,OM,y}$
Data unit	tCO_2/MWh

[21] EB113_repan10_TOOL06_ver04 (unfccc.int)

[22] Microsoft Word - Global-Warming-Potential-Values.docx (ghgprotocol.org)

Description	Operating margin CO_2 emission factor r in year *y*
Source of data	"2023 Operating Margin Emission Factors for Regional Power Grids in China" [23] by DNA of China
Value applied	0.5959 tCO_2/MWh
Justification of choice of data or description of measurement methods and procedures applied	Official and authoritative statistic data.
Purpose of data	Calculation of baseline emissions
Comments	-

Data / Parameter	$EF_{grid,BM,y}$
Data unit	tCO_2/MWh
Description	Build margin CO_2 emission factor
Source of data	"2023 Build Margin Emission Factors for Regional Power Grids in China"[24] by DNA of China
Value applied	0.0634 tCO_2/MWh
Justification of choice of data or description of measurement methods and procedures applied	Official and authoritative statistic data.
Purpose of data	Calculation of baseline emissions
Comments	-

Data / Parameter	w_{OM}
Data unit	%
Description	Weighting of operating margin emissions factor

[23] W020240709709437370462.pdf (ncsc.org.cn)

[24] W020240709709439048191.pdf (ncsc.org.cn)

Source of data	Tool07: "Tool to calculate the emission factor for an electricity system" (version 07.0)
Value applied	50%
Justification of choice of data or description of measurement methods and procedures applied	For projects other than wind and solar power generation project activities, w_{OM} = 0.5 for the first crediting period.
Purpose of data	Calculation of baseline emissions
Comments	-

Data / Parameter	w_{BM}
Data unit	%
Description	Weighting of build margin emissions factor in year *y*
Source of data	Tool07: "Tool to calculate the emission factor for an electricity system" (version 07.0)
Value applied	50%
Justification of choice of data or description of measurement methods and procedures applied	For projects other than wind and solar power generation project activities, w_{BM} = 0.5 for the first crediting period.
Purpose of data	Calculation of baseline emissions
Comments	-

Data / Parameter	$EF_{grid,CM,y}$
Data unit	tCO_2/MWh
Description	Combined margin CO_2 emission factor in year *y*
Source of data	Calculated based on TOOL07: "Tool to calculate the emission factor for an electricity system" (Version 07.0)
Value applied	0.32965 tCO_2/MWh
Justification of choice of data or description of	$EF_{grid,CM,y}$ is calculated based on $EF_{grid,OM,y}$ and $EF_{grid,BM,y}$ as per the latest version of TOOL07

measurement methods and procedures applied	
Purpose of data	Calculation of baseline emissions
Comments	-

Data / Parameter	OX
Data unit	-
Description	Oxidation factor (reflecting the amount of methane from SWDS that is oxidised in the soil or other material covering the waste) (dimensionless)
Source of data	AMS-III.G: "Landfill methane recovery" (Version 10.0)
Value applied	0.1
Justification of choice of data or description of measurement methods and procedures applied	A default value of 0.1 may be used.
Purpose of data	Calculation of baseline emissions
Comments	-

Data / Parameter	η_{PJ}
Data unit	%
Description	Efficiency of the LFG capture system that will be installed in the project activity. It is used for ex ante estimation only.
Source of data	The Project FSR
Value applied	50%
Justification of choice of data or description of measurement methods and procedures applied	A default value of 50% may be used.
Purpose of data	Calculation of baseline emissions
Comments	-

Data / Parameter	$DOC_{f,y}$
Data unit	%
Description	Fraction of degradable organic carbon (DOC) that decomposes under the specific conditions occurring in the SWDS for year *y* (weight fraction)
Source of data	Tool04: "Emissions from solid waste disposal sites" (Version 08.1)
Value applied	50%
Justification of choice of data or description of measurement methods and procedures applied	For methane calculation from MSW, the national value of DOC_f is equal to 0.5.
Purpose of data	Calculation of baseline emissions
Comments	-

Data / Parameter	$W_{j,x}$
Data unit	t
Description	Amount of solid waste type *j* disposed or prevented from disposal in the SWDS in the year *x*
Source of data	Tool04: "Emissions from solid waste disposal sites" (Version 08.1)
Value applied	Refer to the ER sheet
Justification of choice of data or description of measurement methods and procedures applied	It is calculated as total waste amount dumped in the landfill site in the year *x* multiplied by organic waste type *j* fraction on wet basis. Both total waste amount and waste type *j* fraction are referred to FSR.
Purpose of data	Calculation of baseline emissions
Comments	-

Data / Parameter	φ_y
Data unit	-
Description	Model correction factor to account for model uncertainties for year *y*

<table>
<tr><td>Source of data</td><td>Tool04: “Emissions from solid waste disposal sites” (Version 08.1)</td></tr>
<tr><td>Value applied</td><td>0.75</td></tr>
<tr><td>Justification of choice of data or description of measurement methods and procedures applied</td><td>Option 1: use a default value is chosen to calculate φ_y.
<table><tr><td></td><td>Humid/wet conditions</td><td>Dry conditions</td></tr><tr><td>Application A</td><td>0.75</td><td>0.75</td></tr><tr><td>Application B</td><td>0.85</td><td>0.80</td></tr></table>For the Project under Application A: The CDM project activity mitigates methane emissions from a specific existing SWDS, φ_y = 0.75.</td></tr>
<tr><td>Purpose of data</td><td>Calculation of baseline emissions</td></tr>
<tr><td>Comments</td><td>-</td></tr>
</table>

Data / Parameter	f_y
Data unit	%
Description	Fraction of methane captured at the SWDS and flared, combusted or used in another manner that prevents the emissions of methane to the atmosphere in year *y*
Source of data	AMS-III.G Landfill methane recovery (Version 10.0)
Value applied	0
Justification of choice of data or description of measurement methods and procedures applied	With the factor “f=0.0” because the amount of LFG that would have been captured and destroyed is already accounted for in Equation 4. For the Project, the amount of LFG that would have been captured and destroyed is already counted in the parameter $F_{CH4,BL,y}$ in the equation. Hence f_y, = 0.
Purpose of data	Calculation of baseline emissions
Comments	For Application A: f_y is calculated once for the crediting period.

Data / Parameter	F
Data unit	%
Description	Fraction of methane in the SWDS gas (volume fraction)

Source of data	Tool04: "Emissions from solid waste disposal sites" (Version 08.1)
Value applied:	50%
Justification of choice of data or description of measurement methods and procedures applied	The default value from the latest version of Tool04 is 0.5.
Purpose of data	Calculation of baseline emissions
Comments	-

Data / Parameter	MCF_y
Data unit	-
Description	Methane correction factor for year *y*
Source of data	Tool04: "Emissions from solid waste disposal sites" (Version 08.1)
Value applied:	1.0
Justification of choice of data or description of measurement methods and procedures applied	In case of Application A, MCF_y = 1.0 for anaerobic managed solid waste disposal sites. These must have controlled placement of waste (i.e. waste directed to specific deposition areas, a degree of control of scavenging and a degree of control of fires) and will include at least one of the following: (i) cover material; (ii) mechanical compacting; or (iii) levelling of the waste.
Purpose of data	Calculation of baseline emissions
Comments	-

Data / Parameter	DOC_j
Data unit	%
Description	Fraction of degradable organic carbon in the waste type *j* (weight fraction)
Source of data	Tool04: "Emissions from solid waste disposal sites" (Version 08.1)
Value applied:	see table below
Justification of choice of data or description of	Using default value provided by Tool04 (Version 08.1)

Value applied:

j	a	b	c	d	e
	Wood	Paper	Food	Textile	Others (rubber, plastic, glass, metal, etc.)
DOC_j	43%	40%	15%	24%	0%

measurement methods and procedures applied	
Purpose of data	Calculation of baseline emissions
Comments	-

<table>
<tr><td>Data / Parameter</td><td>k_j</td></tr>
<tr><td>Data unit</td><td>1 / yr</td></tr>
<tr><td>Description</td><td>Decay rate for the waste type j</td></tr>
<tr><td>Source of data</td><td>Tool04: “Emissions from solid waste disposal sites” (Version 08.1)</td></tr>
<tr><td>Value applied:</td><td>
<table>
<tr><td rowspan="2">j</td><td>a</td><td>b</td><td>c</td><td>d</td><td>e</td></tr>
<tr><td>Wood</td><td>Paper</td><td>Kitchen</td><td>Textile</td><td>Others (rubber, plastic, glass, metal, etc.)</td></tr>
<tr><td>k_j</td><td>0.03</td><td>0.06</td><td>0.185</td><td>0.06</td><td>0</td></tr>
</table>
</td></tr>
<tr><td>Justification of choice of data or description of measurement methods and procedures applied</td><td>Using following default value for waste type j:
<table>
<tr><td colspan="2" rowspan="2">Waste type j</td><td colspan="2">Boreal and Temperate (MAT ≤20℃)</td><td colspan="2">Tropical (MAT >20℃)</td></tr>
<tr><td>Dry (MAP/PET <1)</td><td>Wet (MAP/PET > 1)</td><td>Dry (MAP<1000 mm)</td><td>Dry (MAP>1000 mm)</td></tr>
<tr><td rowspan="2">Slowly degrading</td><td>Pulp, paper, cardboard (other than sludge), textiles</td><td>0.04</td><td>0.06</td><td>0.045</td><td>0.07</td></tr>
<tr><td>Wood, wood product and straw</td><td>0.02</td><td>0.03</td><td>0.025</td><td>0.035</td></tr>
<tr><td>Moderately degrading</td><td>Other (non-food) organic putrescible garden and park waste</td><td>0.05</td><td>0.10</td><td>0.065</td><td>0.17</td></tr>
<tr><td>Rapidly degrading</td><td>Food, food waste, sewage sludge, beverages and tobacco</td><td>0.06</td><td>0.185</td><td>0.085</td><td>0.40</td></tr>
</table>
</td></tr>
</table>

	Note: MAT – mean annual temperature, MAP – Mean annual precipitation, PET – potential evapotranspiration. MAP/PET is the ratio between the mean annual precipitation and the potential evapotranspiration. For the project site in Fengdu Chongqing, the climate zone is subtropical monsoon humid climate with an annual average temperature is 17.7 ℃[25], below 20℃. The average annual PET in Chongqing city is about 1136.5 mm/yr[26]. The annual average precipitation is about 730 mm/yr[27], thus MAP/PET >1. Therefore, the values for Boreal and Temperate (MAT≤20℃) and Wet (MAP/PET>1) are applicable.
Purpose of data	Calculation of baseline emissions
Comments	-

5.2 Data and Parameters Monitored

Data / Parameter	$EG_{PJ,facility,y}$
Data unit	MWh
Description	Net amount of electricity generated using LFG by the project activity in year *y*
Source of data	Electricity meter
Description of measurement methods and procedures to be applied	Monitored by a bi-directional electricity meter M1 installed at the outlet of the onsite substation. Meter readings will be read and recorded by the project staff monthly. Sales receipts will be issued for electricity settlement monthly.
Frequency of monitoring/recording	Continuously measured and monthly recorded
Value applied	Refer to ER sheet
Monitoring equipment	Electricity meter
QA/QC procedures to be applied	The monitored values will be cross checked with sales receipts.

[25] https://www.cq.gov.cn/zjcq/sqgk/cqsq/202312/t20231231_13501976.html

[26] https://www.cq.gov.cn/zjcq/sqgk/cqsq/202312/t20231231_13501976.html

[27] WANG Yuwen, GAO Yanghua, HU Huailin. Development and utilization of climatic resources in the Three Gorges Reservoir area of Chongqing. Development of mountainous areas[J],2001,(12):12-20.

	The electricity meter is to be calibrated regularly in compliance with the latest version of "*Technical administrative code of electric energy metering*". The calibration reports, recorded data files and sales receipts will be archived for two years following the end of the crediting period.
Purpose of data	Calculation of baseline emissions
Calculation method	This parameter should be calculated as difference between (a) the quantity of electricity supplied by the project to the grid ($\mathrm{EG}_{\mathrm{export,y}}$); and (b) the quantity of electricity the project from the grid ($\mathrm{EG}_{\mathrm{import,y}}$). The formula is as follows: $EG_{PJ,y} = EG_{export,y} - EG_{import,y}$ $EG_{PJ,facility,y} = EG_{PJy}$ The following parameters shall be measured: (a) The quantity of electricity supplied by the project to the grid; and (b) The quantity of electricity delivered to the project from the grid
Comments	This parameter is required for calculating baseline emissions associated with electricity generation ($BE_{EC,y}$) using the methodological tool "Baseline, project and/or leakage emissions from electricity consumption and monitoring of electricity generation" (Version 18.0).

Data / Parameter	EG_y
Data unit	MWh
Description	Electricity generation by the project activity in year *y*
Source of data	Electricity meter
Description of measurement methods and procedures to be applied	Measured by a bi-directional electricity meter M2 installed at the outlet of gas generators. Meter readings will be read and recorded by the project staff monthly.
Frequency of monitoring/recording	Continuously measured and regularly recorded
Value applied	Ex post calculation only
Monitoring equipment	Electricity meter
QA/QC procedures to be applied	The electricity meter is to be calibrated regularly in compliance with the latest version of "Technical administrative code of electric energy metering".

Purpose of data	Calculation of emission reduction
Calculation method	-
Comments	This parameter is required for calculating baseline emissions

5.3 Monitoring Plan

The monitoring plan presented in this PD assures that real, measurable, long-term GHG emission reductions can be monitored, recorded and reported. It is a crucial procedure to identify the final VCUs of the project. This monitoring plan will be implemented by the project owner during the project operation period. The details of the monitoring plan are specified as follows.

The Project owner organizes a specific VCS team in project development department to be responsible for data collection, supervision and witness the whole process of data measuring and recording.

(A) Management structure

The Project owner organizes a specific VCS team in project development department to be responsible for data collection, supervision and witness the whole process of data measuring and recording. A VCS manager is appointed to take full responsibility for the overall monitoring of the project. The monitoring and measurement of extracted gas, electricity generation and consumption etc. are carried out by a few designated monitoring officers. In addition, the Project developer appoints internal verifiers who is responsible for internal check of the measurement, collection of relevant receipts and invoices, and the calculation of the emission reductions. A monitoring and management manual of the project that identifies detailed duties and responsibilities of the relevant parties is developed and served as the basis of the project monitoring. Figure 5.1 shows the operation and management structure of the Project.

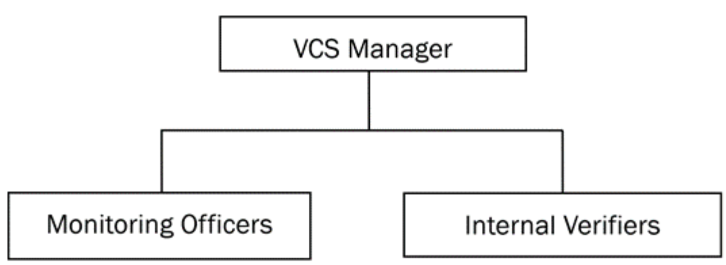

Figure 5.1 Operation and management structure of the project

(B) Data and Parameters to be monitored

Figure 5.2 shows the parameters to be monitored as well as the positions of the monitoring instruments; Table 5.1 further explains each of the monitored parameters.

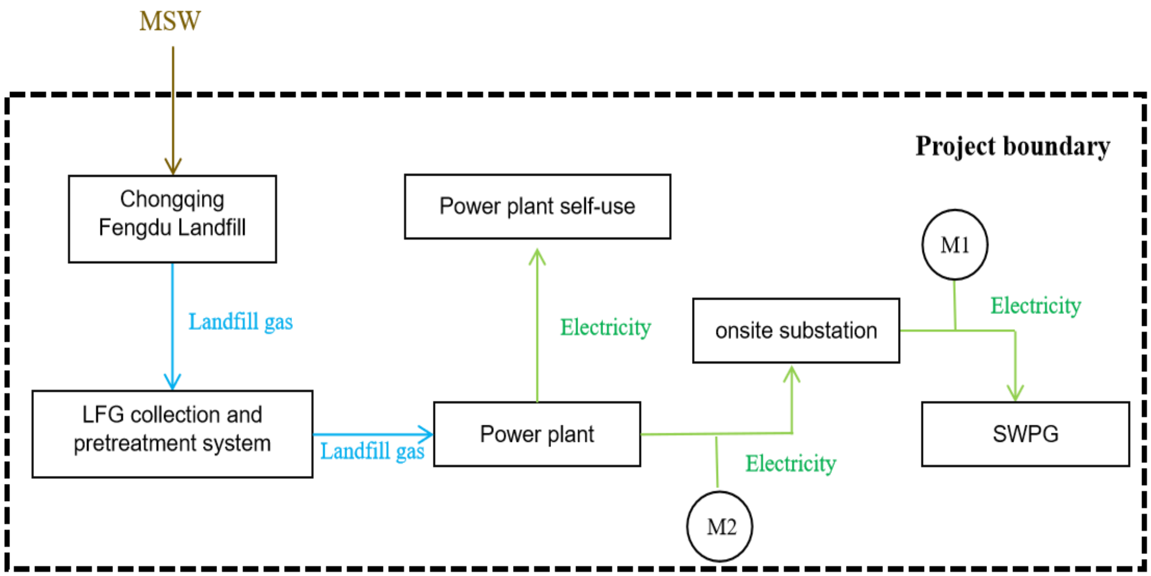

Figure 5.2 Project monitoring diagram

Table 5.1 Description of the monitored parameters

Position	Monitoring instrument	Monitored parameter(s)	Description
M1	Electricity meter	$EG_{PJ,facility,y}$ ($EG_{PJ,y}$)	Quantity of net electricity generation supplied by the project to the grid is continuously monitored by bi-directional electricity meter installed by outlet of onsite substation with the accuracy of ±1%.
M2	Electricity meter	EG_y	Amount of electricity generated using LFG by the project activity is continuously monitored by electricity meter installed at the outlet of gas generator with the accuracy of ±1%.

(C) Data collection

Monitoring officers are responsible for data collection. Designated teams will read and collect the monitored data regularly. The computer system will automatically monitor and record relevant meter data. Automatic records will serve as the main data source for emission reductions calculation. All data files, relevant purchase invoices and sales receipts will be collected by a designated monitoring officer, who will prepare backup in time and archive all documents properly.

(D) Quality assurance

All the monitoring instruments are chosen in compliance with VCS requirements, and will be calibrated regularly for accuracy by qualified third-party organizations according to the national regulations during the entire crediting period. To assist in future verifications, the project owner preserves the calibration records, along with the files of monitored data.

Error-checking is carried out regularly on site and at the point of data storage to detect data measuring/transmission failures as well as malfunctions. In case of malfunction of any measuring instrument, the instrument supplier will provide technical support to solve the problem promptly, and the emission reductions during the troubleshooting period will be calculated conservatively.

The installation of the electricity metering equipment will fulfill the requirements of “DL/T448-2016 Power Metering Device Technical Management Procedures”. The accuracy of all metering equipment will fulfill the relevant national standard. All metering equipment will be calibrated regularly for accuracy by qualified party according to the national regulations.

(E) Data file management

All the monitored data are electronically files at the end of each month and the electronic data files are archived in both disk copy and printed hard copy. All data collected as part of the monitoring must be archived electronically and be kept at least for two years after the end of the crediting period. The project owner will provide original records and documents if necessary.

(F) Emergency

The monitoring team members are in charge of identifying any emergency, such as failure or malfunction of a monitoring instrument, and then reporting it to the VCS Manager. The VCS Manager will promptly turn to professionals, such as technicians/engineers from the plant and/or equipment suppliers, to address the emergency. When the issue is successfully handled, the manager will inform the monitoring team members that the emergency is ended.

APPENDIX 1: COMMERCIALLY SENSITIVE INFORMATION

The project does not include commercially sensitive information in the project description to be excluded in the public version.

Section	Information	Justification
N/A	N/A	N/A

APPENDIX 2: CALCULATION OF BASELINE EMISSION FACTOR FOR SWPG

Evidence of $EF_{grid,OM,y}$ calculation:

According to "Tool to calculate the emission factor for an electricity system" (version 07.0), the following six steps are applied to calculate the project baseline emission factor:

(a) **Step 1:** Identify the relevant electricity systems

Project participants may delineate the project electricity system using any of the following options:

- **Option 1.** A delineation of the project electricity system and connected electricity systems published by the DNA or the group of the DNAs of the host country(ies). In case a delineation is provided by a group of DNAs, the same delineation should be used by all the project participants applying the tool in these countries;
- **Option 2.** A delineation of the project electricity system defined by the dispatch area of the dispatch centre responsible for scheduling and dispatching electricity generated by the project activity. Where the dispatch area is controlled by more than one dispatch centre, i.e. layered dispatch area, the higher level area shall be used as a delineation of the project electricity system (e.g. where regional dispatch centres are required to comply with dispatch orders of the national dispatch centre then area controlled by the national dispatch centre shall be used);
- **Option 3.** A delineation of the project electricity system defined by more than one independent dispatch areas, e.g. multi-national power pools.

The Chinese DNA has published a delineation of the project's electricity system and connected electricity systems, Option 1 is applied for the project. According to the delineations, the Central China Power Grid (CCPG) is identified as the relevant electric power system of the project, which includes Henan Province, Hubei Province, Hunan Province, Jiangxi Province, Sichuan Province, and Chongqing City.

(b) **Step 2:** Choose whether to include off-grid power plants in the project electricity system (optional)

Project participants may choose between the following two options to calculate the operating margin and build margin emission factor:

- **Optional I:** Only grid power plants are included in the calculation.
- **Optional II:** Both grid power plants and off-grid power plants are included in the calculation.

Based on China's real situation, only grid power plants are included in the calculation.

(c) **Step 3:** Select a method to determine the operating margin (OM)

The calculation of the operating margin emission factor ($EF_{grid,OM,y}$) is based on one of the following methods, which are described under Step 4:

- Simple OM; or
- Simple adjusted OM; or
- Dispatch data analysis OM; or
- Average OM.

According to the data from Electric Power Industry Statistical Data Compilation 2017 – 2021 and China Energy Statistical Yearbook 2018 – 2022, from 2017 to 2021, for the SWPG the project activity connected to, the low-cost/must-run resources (LCMR)[28] electric power resources generation accounting for the total grid total is 81.78%, 79.64%, 79.14%, 79.38%, and 76.99%, respectively, all higher than 50%, which can not satisfied the applicability of the method (a), but the adjusted simple OM method is applied for the calculation of the operating margin emission factor .

On the grounds of Flow chart: Overview of the application of OM methods in “Tool to calculate the emission factor for an electricity system” (version 07.0) as shown in Figure 0-1, since the LCMR share is less than 50% of total grid generation in the average of the five most recent years, the simple OM method is chosen for the calculation of the OM emission factor EF.

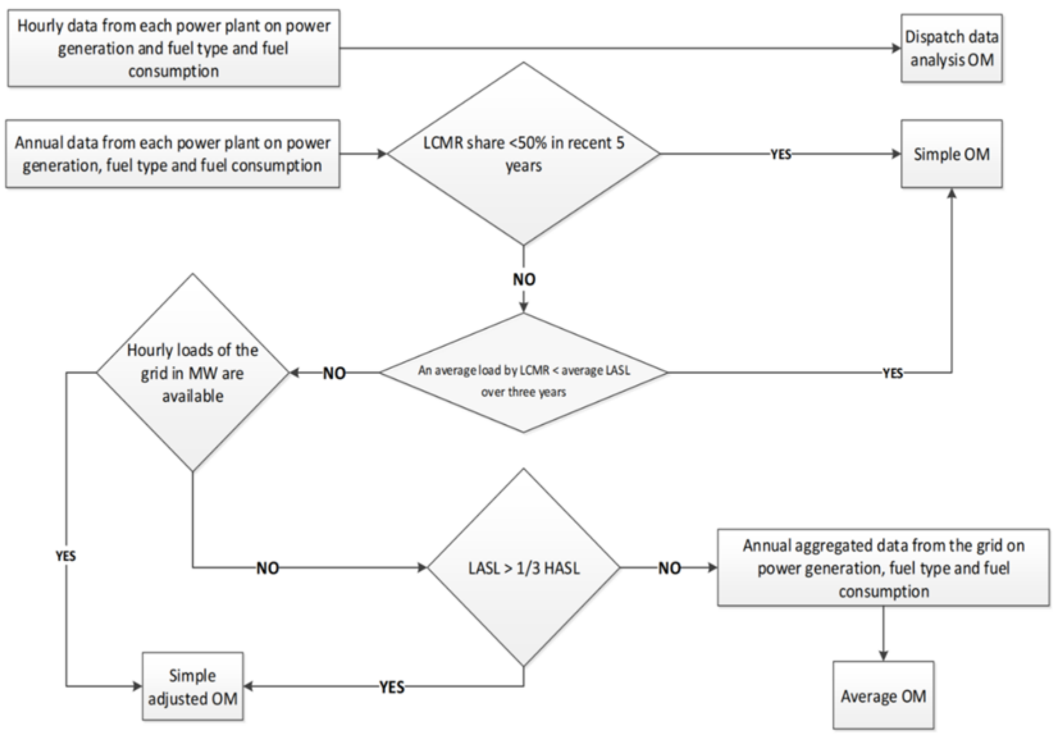

Figure 0 1 Flow chart: Overview of the application of OM methods

[28] Low-cost/must-run resources are defined as power plants with low marginal generation costs or power plants that are dispatched independently of the daily or seasonal load of the grid. They typically include hydro, geothermal, wind, low-cost biomass, nuclear and solar generation. If coal is obviously used as must-run, it should also be included in this list.

As per the latest version of "Tool to calculate the emission factor for an electricity system" (version 07.0), the emissions factor for the simple OM can be calculated using either of the two following data vintages:

- **Ex ante option:** if the ex ante option is chosen, the emission factor is determined once at the validation stage, thus no monitoring and recalculation of the emissions factor during the crediting period is required. For grid power plants, use a 3-year generation-weighted average, based on the most recent data available at the time of submission of the CDM-PDD to the DOE for validation. For off-grid power plants, use a single calendar year within the five most recent calendar years prior to the time of submission of the CDM-PDD for validation;
- **Ex post option:** if the ex post option is chosen, the emission factor is determined for the year in which the project activity displaces grid electricity, requiring the emissions factor to be updated annually during monitoring. If the data required to calculate the emission factor for year y is usually only available later than six months after the end of year y, alternatively the emission factor of the previous year y-1 may be used. If the data is usually only available 18 months after the end of year y, the emission factor of the year proceeding the previous year y-2 may be used. The same data vintage (y, y-1 or y-2) should be used throughout all crediting periods.

The Ex ante option is selected, and the $EF_{grid,OM}$ is fixed during the crediting period.

(d) **Step 4:** Calculate the operating margin emission factor according to the selected method

Since the simple OM method is determined to calculate the CCPG OM emission factor in step 3, the simple OM emission factor is calculated as the generation-weighted average CO emissions per unit net electricity generation (t CO_2/MWh) of all generating power plants

serving the system, not including low-cost/must-run power plants/units.

The simple OM may be calculated by one of the following two options:

- **Option A**: Based on the net electricity generation and a CO_2 emission factor of each power unit; or
- **Option B**: Based on the total net electricity generation of all power plants serving the system and the fuel types and total fuel consumption of the project electricity system. Option B can only be used if:
  - (i) The necessary data for Option A is not available; and
  - (ii) Only nuclear and renewable power generation are considered as low-cost/must-run power sources and the quantity of electricity supplied to the grid by these sources is known; and
  - (iii) Off-grid power plants are not included in the calculation (i.e. if Option I has been chosen in Step 2).

For the project activity, the required data for exercising Option A are unavailable, and those of Option B can be obtained from official sources. Off-grid power plants are not included in the calculation; therefore, Option B is chosen to calculate the operating margin emission factor.

Under Option B: Calculation based on total fuel consumption and electricity generation of the system, The simple OM emission factor is calculated based on the net electricity supplied to the grid by all power plants serving the system, not including low-cost/must-run power plants/units, and based on the fuel type(s) and total fuel consumption of the project electricity system, as follows:

$$EG_{grid,OMsimple,y} = \frac{\sum_i FC_{i,y} \times NCV_{i,y} \times EF_{CO2,i,y}}{EG_y} \quad \text{Equation 1}$$

Where:

$EF_{grid,OMsimple,y}$	=	Simple operating margin CO_2 emission factor in year *y* (t CO_2/MWh)
$FC_{i,y}$	=	Amount of fuel type *i* consumed in the project electricity system in year *y* (mass or volume unit)
$NCV_{i,y}$	=	Net calorific value (energy content) of fuel type *i* in year *y* (GJ/mass or volume unit)
$EF_{CO2,i,y}$	=	CO_2 emission factor of fuel type *i* in year *y* (t CO_2/GJ)
EG_y	=	Net electricity generated and delivered to the grid by all power sources serving the system, not including low-cost/must-run power plants/units, in year *y* (MWh)
i	=	All fuel types combusted in power sources in the project electricity system in year *y*
y	=	The relevant year as per the data vintage chosen in Step 3

For this approach (simple OM) to calculate the operating margin, the subscript *m* refers to the power plants/units delivering electricity to the grid, not including low-cost/must-run power plants/units.

Data such as EG_y, $FC_{i,y}$ and $NCV_{i,y}$ used in the calculation of OM are derived from the China Energy Statistical Yearbook 2020 -- 2022 and the Statistical Survey System of Energy Consumption of Public Institutions (formulated by the National Government Offices Administration and approved by the National Bureau of Statistics, Aug 2019). The data of plant electricity consumption rate are derived from the China Electricity Yearbook 2020 -- 2022, the data of electricity exchange between grids are derived from the Statistical Data of the Electric Power Industry 2019 -- 2021, and $EF_{CO2,i,y}$ are derived from Table 1.4, Chapter 1 of the Energy volume of the 2006 IPCC Guidelines for the Preparation of National Inventories. The lower limit of 95% confidence interval for each fuel emission factor is obtained according to the conservative principle.

Based on these data, the latest Simple OM Emission Factor ($EF_{grid,OMsimple,y}$) of the SWPG is 0.5959 tCO_2/MWh in 2023[29].

(e) **Step 5:** Calculate the build margin (BM) emission factor

In terms of vintage of data, project participants can choose between one of the following two options:

- **Option 1**: for the first crediting period, calculate the build margin emission factor ex ante based on the most recent information available on units already built for sample group m at the time of CDM-PDD submission to the DOE for validation. For the second crediting period, the build margin emission factor should be updated based on the most recent information available on units already built at the time of submission of the request for renewal of the crediting period to the DOE. For the third crediting period, the build margin emission factor calculated for the second crediting period should be used. This option does not require monitoring the emission factor during the crediting period;
- **Option 2**: For the first crediting period, the build margin emission factor shall be updated annually, ex post, including those units built up to the year of registration of the project activity or, if information up to the year of registration is not yet available, including those units built up to the latest year for which information is available. For the second crediting period, the build margin emissions factor shall be calculated ex ante, as described in Option 1 above. For the third crediting period, the build margin emission factor calculated for the second crediting period should be used.

For the Project, Option 1 is chosen to calculate the ex ante BM emission factor based on the most recent information available on units already built for sample group m at the time of this project description submission.

The sample group of power units *m* used to calculate the build margin should be determined as per the following procedure, consistent with the data vintage selected above:

(i) Identify the set of five power units, excluding power units registered as CDM project activities, that started to supply electricity to the grid most recently ($SET_{5\text{-}units}$) and determine their annual electricity generation ($AEG_{SET\text{-}5\text{-}units}$, in MWh);

(ii) Determine the annual electricity generation of the project electricity system, excluding power units registered as CDM project activities (AEG_{total}, in MWh). Identify the set of power units, excluding power units registered as CDM project activities, that started to supply electricity to the grid most recently and that comprise 20 per cent of AEG_{total} (if 20 per cent falls on part of the generation of a unit, the generation of that unit is fully included in the calculation) ($SET_{\geq 20\ per\ cent}$) and determine their annual electricity generation ($AEG_{SET\text{-}\geq 20\ per\ cent}$, in MWh);

(iii) From $SET_{5\text{-}units}$ and $SET_{\geq 20\ per\ cent}$ select the set of power units that comprises the larger annual electricity generation (SET_{sample}); Identify the date when the power units in SET_{sample} started to supply electricity to the grid. If none of the power units in SET_{sample}

[29] W020240709709437370462.pdf (ncsc.org.cn)

started to supply electricity to the grid more than 10 years ago, then use SET_{sample} to calculate the build margin. In this case ignore Steps (iv), (v) and (vi).

Otherwise:

(iv) Exclude from SET_{sample} the power units which started to supply electricity to the grid more than 10 years ago. Include in that set the power units registered as CDM project activities, starting with power units that started to supply electricity to the grid most recently, until the electricity generation of the new set comprises 20 per cent of the annual electricity generation of the project electricity system (if 20 per cent falls on part of the generation of a unit, the generation of that unit is fully included in the calculation) to the extent is possible. Determine for the resulting set ($SET_{sample\text{-}CDM}$) the annual electricity generation ($AEG_{SET\text{-}sample\text{-}CDM}$, in MWh);

If the annual electricity generation of that set is comprises at least 20 per cent of the annual electricity generation of the project electricity system (i.e. $AEG_{SET\text{-}sample\text{-}CDM} \geq 0.2 \times AEG_{total}$), then use the sample group $SET_{sample\text{-}CDM}$ to calculate the build margin. Ignore Steps (v) and (vi)

Otherwise:

(v) Include in the sample group $SET_{sample\text{-}CDM}$ the power units that started to supply electricity to the grid more than 10 years ago until the electricity generation of the new set comprises 20 per cent of the annual electricity generation of the project electricity system (if 20 per cent falls on part of the generation of a unit, the generation of that unit is fully included in the calculation);

(vi) The sample group of power units m used to calculate the build margin is the resulting set ($SET_{sample\text{-}CDM\text{-}>10yrs}$)

The build margin emissions factor is the generation-weighted average emission factor (t CO_2/MWh) of all power units m during the most recent year y for which electricity generation data is available, calculated as follows:

$$EF_{grid,BM,y} = \frac{\sum_m EG_{m,y} \times EF_{EL,m,y}}{\sum_m EG_{m,y}} \quad \text{Equation 2}$$

Where:

$EF_{grid,BM,y}$	=	Build margin CO_2 emission factor in year y (t CO_2/MWh)
$EG_{m,y}$	=	Net quantity of electricity generated and delivered to the grid by power unit m in year y (mass or volume unit)
$EF_{EL,m,y}$	=	CO_2 emission factor of power unit m in year y (t CO_2/GJ)
m	=	Power units included in the build margin
y	=	Most recent historical year for which electricity generation data is available

According to the instructions of China DNA[30], for the determination of the set of samples, a sample merging processing in some degree has been adopted due to that the power generation data, energy consumption data or thermal efficiency data of each plant cannot be consulted in the public statistical data. In this calculation, the newly-installed power units in the past years are classified by year, province and power generation technology, and the same type of newly-installed power units in the same province and in the same year are bundled as a "newly-installed power units".

The power generation of each "newly-installed power units" in the most recent year y is estimated based on its installed capacity and the number of power generation utilization hours in year y. The formula is as follows:

$$EG_{m,y} = CAP_m \times H_{m,y} \qquad \text{Equation 3}$$

Where:

$EG_{m,y}$	=	Net quantity of electricity generated and delivered to the grid by power unit *m* in year *y* (MWh)
CAP_m	=	Installed capacity of electricity generated and delivered to the grid by power unit *m* in year *y* (MW)
$H_{m,y}$	=	The number of power utilization hours (h) of electricity generated and delivered to the grid by power unit *m* in year y. And it selects the average utilization hours of similar units in the province in which it is located in year *y*;
m	=	Power units included in the build margin
y	=	Most recent historical year for which electricity generation data is available

The power unit m is selected from the "newly-installed power plants" in the most recent year *y* (For the calculation of the grid BM in 2023, the *y* is equal to 2021) to the "newly-installed power plants" in the earlier year, until the cumulative power generation reaches 20% of the total power generation in the year *y* (*y* = 2021).

Since the newly-installed power units of the same type (k) in the same province (A) and the same year (t) are bundled into the "newly-installed power units", the CAP_m is equal to the statistical data of recent installed capacity of a given unit type(k) in a given year(t) in a given province (A):

$$CAP_m = CAP_m|_{m=(A,t,k)} = CAP_{A,t,k} \qquad \text{Equation 4}$$

Where

[30] W020240709709436742308.pdf (ncsc.org.cn)

CAP_m	=	Installed capacity of electricity generated and delivered to the grid by power unit *m* in year *y* (MW), and *m* is equivalent to an established combination of (A, t, k)
$CAP_{A,t,k}$	=	Capacity of newly-installed power units of a given province (A), given year (t), and given unit type (k) (MW)
A	=	It is the various provincial regions covered by the regional power grid
t	=	It is the sampling year of the "newly-installed power units". For the calculation of the grid BM in 2023, t is equal to 2021, 2020, until the units that comprise at least 20 percent of the system generation in 2021.
k	=	It is the power generation technology classification of "newly-installed power units", which is divided into hydro-power, coal-thermal power, gas-thermal power, oil-thermal power, Waste-thermal power plant, other thermal power, nuclear power, wind power, solar power, and others.

The CO_2 emission factor of each power unit *m* ($EF_{EL,m,y}$) should be determined as per the guidance in Step 4 section 6.4.1 for the simple OM, using Options A1, A2 or A3, using for *y* the most recent historical year for which electricity generation data is available, and using for *m* the power units included in the build margin.

The emission factor of each power unit *m* should be determined as follows:

- **Option A1:** If for a power unit *m* data on fuel consumption and electricity generation is available, the emission factor ($EF_{EL,m,y}$) should be determined as follows:

$$EF_{EL,m,y} = \frac{\sum_i FC_{i,m,y} \times NCV_{i,y} \times EF_{CO2,i,y}}{EG_{m,y}} \qquad \text{Equation 5}$$

Where:

$EF_{EL,m,y}$	=	CO_2 emission factor of power unit *m* in year *y* (t CO_2/MWh)
$FC_{i,m,y}$	=	Amount of fuel type *i* consumed in the project electricity system in year *y* (mass or volume unit)
$NCV_{i,y}$	=	Net calorific value (energy content) of fuel type *i* in year *y* (GJ/mass or volume unit)
$EF_{CO2,i,y}$	=	CO_2 emission factor of fuel type *i* in year *y* (t CO_2/GJ)

$EG_{m,y}$ = Net quantity of electricity generated and delivered to the grid by power unit *m* in year *y* (MWh)

m = All power units serving the grid in year *y* except low-cost/must-run power units

i = All fuel types combusted in power unit *m* in year *y*

y = The relevant year as per the data vintage chosen in Step 3

- **Option A2:** If for a power unit *m* only data on electricity generation and the fuel types used is available, the emission factor should be determined based on the CO_2 emission factor of the fuel type used and the efficiency of the power unit, as follows:

$$EG_{EL,m,y} = \frac{EF_{CO2,m,i,y} \times 3.6}{\eta_{m,y}} \quad \text{Equation 6}$$

Where:

$EF_{EL,m,y}$ = CO_2 emission factor of power unit *m* in year *y* (t CO_2/MWh)

$EF_{CO2,m,i,y}$ = CO_2 emission factor of fuel type *i* in year *y* (t CO_2/GJ)

$\eta_{m,y}$ = Average net energy conversion efficiency of power unit *m* in year *y* (ratio)

m = All power units serving the grid in year *y* except low-cost/must-run power units

y = The relevant year as per the data vintage chosen in Step 3

3.6 = Conversion factor (GJ/MWh)

Where several fuel types are used in the power unit, use the fuel type with the lowest CO_2 emission factor for $EF_{CO2,m,i,y}$.

- **Option A3:** If for a power unit *m* only data on electricity generation is available, an emission factor of 0 t CO_2/MWh can be assumed as a simple and conservative approach.

Option A2 is selected to determine $EG_{EL,m,y}$ because only data on electricity generation and the fuel types used is available.

If the power units included in the build margin *m* correspond to the sample group $SET_{sample\text{-}CDM\text{->}10yrs}$, then, as a conservative approach, only Option A2 from guidance in Step 4 section 6.4.1 can be used and the default values provided in Table 2, Appendix of TOOL09: "Determining the baseline efficiency of thermal or electric energy generation systems" shall be used to determine the parameter $\eta_{m,y}$ for the power units that started to supply electricity to the grid more than 10 years ago.

The data of $CAP_{A,t,k}$ and $H_{m,y}$ used in the calculation of BM over the past years are mainly derived from the China Electricity Yearbook for 2017-2022, and the data of $\eta_{m,y}$ are derived from the Statistical Data Collection of Electric Power Industry in 2019.

Based on these data, the latest Simple BM Emission Factor ($EF_{grid,BM,y}$) of the SWPG is 0.0634 tCO_2/MWh in 2023[31].

(f) **Step 6:** Calculate the combined margin (CM) emission factor

The calculation of the combined margin (CM) emission factor ($EF_{grid,CM,y}$) is based on one of the following methods:

- **Option a:** Weighted average CM; or
- **Option b:** Simplified CM

According to the Flow chart: Determination of CM emission factor in Figure 0-2, the weighted average CM method is used since data to determine OM and BM are available, which also should be used as the preferred option.

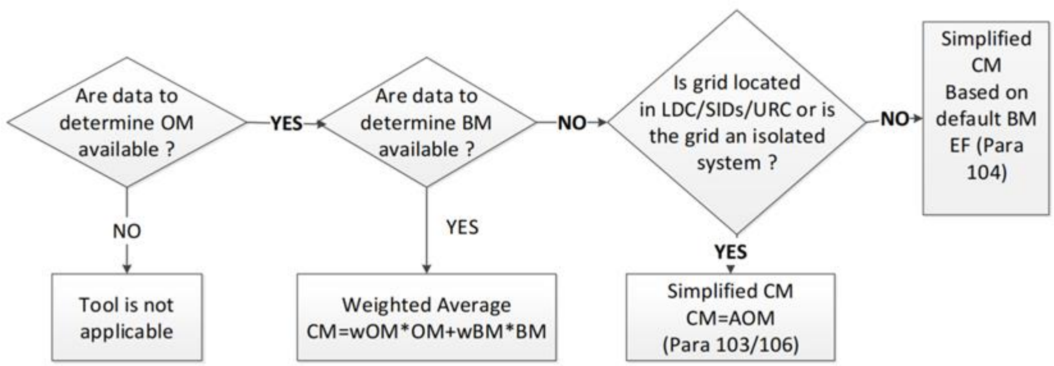

Figure 0 2 Flow chart: Determination of CM emission factor

The combined margin emissions factor is calculated as follows:

$$EF_{grid,OM,y} = EF_{grid,CM,y} \times w_{OM} + EF_{grid,BM,y} \times w_{BM} \quad \text{Equation 7}$$

Where:

$EF_{grid,BM,y}$	=	Build margin CO_2 emission factor in year *y* (t CO_2/MWh)
$EF_{grid,OM,y}$	=	Operating margin CO_2 emission factor in year *y* (t CO_2/MWh)
w_{OM}	=	Weighting of operating margin emissions factor (per cent)
w_{BM}	=	Weighting of build margin emissions factor (per cent)

[31] W020240709709439048191.pdf (ncsc.org.cn)

The following default values should be used for w_{OM} and w_{BM}:

- Wind and solar power generation project activities: w_{OM} = 0.75 and w_{BM} = 0.25 (owing to their intermittent and non-dispatchable nature) for the first crediting period and for subsequent crediting periods;
- All other projects: w_{OM} = 0.5 and w_{BM} = 0.5 for the first crediting period, and w_{OM} = 0.25 and w_{BM} = 0.75 for the second and third crediting period, unless otherwise specified in the approved methodology which refers to this tool.

Considering that the Project is a LFG power generation project instead of wind and solar power generation projects, w_{OM} = 0.5 and w_{BM} = 0.5 for the first crediting period, and w_{OM} = 0.25 and w_{BM} = 0.75 for the second and third crediting period. Thus,

$$EF_{grid,OM,y} = 0.5959\, tCO_2/MWh \times 0.5 + 0.0634\, tCO_2/MWh \times 0.5 = 0.32965\, tCO_2/MWh$$